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Differentiate football tactical styles using relevant football analytics metrics, relying on machine learning

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Disclaimer

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Geneva, August 25th, 2025

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Abstract

We aimed to develop an unsupervised clustering model using machine learning to group football tactical styles into groups of teams. The goal was to discover how teams in five different competitions play tactically and what their various styles of play are. The tactical styles of teams were determined using mathematical formulations, logical reasoning, and metrics to analyze teams' tactical playing styles, which would reduce subjectivity and personal opinions and apply more math-based outcomes.

We analyzed data from five European competitions (Premier League, Bundesliga, La Liga (then known as Liga BBVA), Serie A, and Ligue 1) of the 2015/2016 season using the open-access StatsBomb dataset. We used football metrics as features in our models to reveal tactical playing styles across various teams. In total, 23 metrics were used as features that would give us a diverse range of tactical playing styles, including pressing behavior, transition efficiency, building-up phases, defensive and attacking transitions, wing and central play, individual play, and a lot more.

Principal Component Analysis (PCA) was used to reduce dimensionality and better control correlated metrics, followed by five different clustering algorithms to group teams into tactical identity. After testing different types of metrics as features in my model and comparing them with external sources on how these teams played in the 2015/2016 season. I was able to find the ideal model, which was different for every competition. For example, in the Premier League, it was k-means with 9 clusters, but in Ligue 1, it was k-means with 6 clusters. These clusters provide a data-driven perspective of tactical diversities in football.

The models revealed how uniquely some teams played tactically in that season, like Leicester City in that 2015/2016 season, which was clustered alone and was far away from all other teams, indicating a unique style of play. To understand how special that season was in the Premier League, the last time a football team not among the "big four" or the "big six" in the Premier League (OneFootball, 2025) won the trophy was in the 1994/1995 season, which was 20 years earlier. Also, Leicester City had one of the lowest total squad costs in the Premier League, with only the promoted teams ranking below them (The Independent, 2016). Other teams that were clustered alone and played unique tactics in that season were also Barcelona, Real Madrid, Paris Saint-Germain, Bayern Munich, and many more.

Table of contents

Disclaimer 1
Acknowledgments 2
Abstract 3
Table of contents 4
List of tables 7
List of figures 7
Note 8
List of Abbreviations 9
1. Introduction 10
2. Literature review 12
2.1 The importance of playing styles in football 12
2.2 Coaches and football tactical styles 12
2.3 Characterizing tactical playing styles in football 13
2.3.1 The 5 key phases of a football game 13
2.3.2 Exploratory Factor Analysis (EFA) 14
2.3.3 Football analytical tools and metrics 16
2.4 Identifying tactical styles in football using machine learning 16
2.4.1 Spatio-temporal analysis using tracking data 16
2.4.2 Random forest and decision trees 16
2.4.3 Unsupervised clustering of tactical styles 17
2.4.3.1 Topic modeling 17
2.4.3.2 Graph-based 17
2.4.3.3 A model proposing 19 playing styles 18
2.4.4 Supervised classification of tactical styles 18
2.4.4.1 Bag of Words method 19
2.4.5 Limitations 19
2.4.5.1 Limits of unsupervised clustering 19
2.4.5.2 Limits of supervised classification 19
2.5 In summary 20
2.6 Research direction 20
3. Data types for football analytics 21
3.1 Event data 21
3.2 Tracking data 22
4. Methodology 23
4.1 Data source used 23
4.2 Data structure 23
4.3 Selected and designed metrics 25
4.3.1 Transforming event data into SPADL format 25

4.3.2 VAEP metrics 26
4.3.3 From SPADL to tactical features 27
4.3.4 Other metrics beyond VAEP 27
4.3.5 EFA-related indicators 28
4.3.6 The 19 playing styles adapted 28
4.4 Tactical feature selection and model construction 30
4.5 Clustering techniques 31
4.5.1 Evaluation metrics 32
4.5.2 Number of cluster configurations 32
4.6 Choosing the best models per competition 34
4.6.1 Stage 1 35
4.6.2 Stage 2 37
4.6.3 Choosing the best global model 41
4.7 Dimensionality reduction with PCA 41
5. Results 43
5.1 PCA interpretation for every league 43
5.2 League-specific best models 44
5.2.1 Premier League 44
5.2.2 Bundesliga 45
5.2.3 La Liga 47
5.2.4 Serie A 48
5.2.5 Ligue 1 50
5.2.6 Global model 51
6. Discussion 52
6.1 Tactical alignment 52
6.2 Premier League known tactical styles of play 52
6.3 Bundesliga's known tactical styles of play 53
6.4 Serie A known tactical styles of play 54
6.5 La Liga's known tactical styles of play 55
6.6 Ligue 1 known tactical styles of play 56
6.7 Conclusion regarding all competitions 56
6.8 Limitations 57
6.8.1 One unified model 57
6.8.2 Tracking data 57
6.8.3 Metrics 57
6.8.4 Concrete sources 58
6.9 Future work 58
Bibliography 59

Appendix 1: Full table of scores of every team per metric in the Premier League 65

Appendix 2: Associating styles of play for every cluster based on the metric scores for every league 67

Appendix 3: GitHub code link 72

List of tables

Table 1: The five moments of play (source from (Hewitt, Greenham, Norton 2016))....13
Table 2: The 6 new factors identifying playing styles (source from (Fernandez-Navarro et al. 2016))....15
Table 3: Attributing metrics for every style....29
Table 4: Steps for stage 1....35
Table 5: Explanation of the colors and the signs....38
Table 6: Associating styles of play for every cluster based on the metric scores (see Appendix 2 for all the other leagues)....39
Table 7: Results of PCA for every competition....43

List of figures

Figure 1: Example of what the Competitions.json file contains....23
Figure 2: Example event data for a specific season....24
Figure 3: Example of event data for a specific match....25
Figure 4: Premier League evaluation metrics per clustering algorithm....33
Figure 5: Bundesliga evaluation metrics per clustering algorithm....34
Figure 6: Evaluation metrics for k-means for the Premier League....36
Figure 7: Final ranking of the best models per clustering algorithm....37
Figure 8: Scores of every team per metric.....38
Figure 9: K-means clustering for the Premier League....39
Figure 10: 2D visualization of the Premier League....44
Figure 11: 2D visualization of the Bundesliga....46
Figure 12: 2D visualization of La Liga....47
Figure 13: 2D visualization of Serie A....49
Figure 14: 2D visualization of Ligue 1....50
Figure 16: Premier League top models....51
Figure 15: La Liga top models....51
Figure 17: Bundesliga top models....51
Figure 18: Serie A top models....51
Figure 19: Ligue 1 top models....51

Note

This Bachelor thesis used the assistance of generative AI tools such as ChatGPT and Claude AI for most parts of the code. It was used to help me calculate metrics, train unsupervised clustering algorithms using Principal Component Analysis (PCA), and also to debug errors. It was also used for some reformulations of sentences to improve the overall quality of the thesis while maintaining the originality of the work.

Here in the following entry (Hoti 2025), you can find all the code that I used for my work.

List of Abbreviations

HEG	Haute École de Gestion de Genève
PPDA	Passes Allowed Per Defensive Action
VAEP	Valuing Actions by Estimating Probabilities
AI	Artificial Intelligence
PCA	Principal Component Analysis
EFA	Exploratory Factor Analysis
SPADL	Soccer Player Action Description Language
xG	Expected Goals
T3M	Team Tactic Topic Model
LPM	Local Positioning Measurement
XGBoost	Extreme Gradient Boosting
GMM	Gaussian Mixture Model
BIRCH	Balanced Iterative Reducing and Clustering using Hierarchies

1. Introduction

As of today, there is no universally accepted definition of the different football tactical styles that exist in modern football. Analysts and professionals have studied and written articles about different football tactical styles that exist, but most of the time, these definitions are subjective opinions. Of course, some are more obvious than others, like when a team conserves the ball most of the time in every football game. We can presume that this team is playing what is called “possession-based” football, where they dominate possession of the ball throughout the whole game. But things start to get complicated when a specific team changes its tactical style throughout a season or develops a new playing style from season to season. For example, a well-known football club like Liverpool FC, under Jurgen Klopp, used to play in different tactical styles throughout his years. First, they started playing a style called “counter-pressing”, where players play higher on the pitch to regain the ball from the opponent, and then, in the following years, he switched to a more “transition attack”, where Liverpool had more control of the ball on the pitch. At the end of his career in Liverpool, Klopp played a more controlled “possession-based” football where the players were much more patient on going forward to attack, preferring to build up the attack slowly (Coaches’ Voice 2024).

Not only do managers change styles of play, but they also play different ones at the same time. This is needed to adjust some minor problems in the pitch to gain more advantage from the opponent or to counter their style of play. For example, Pep Guardiola at Manchester City sometimes used to play 2, 3, or 4 styles of play in a single football match (Høgli 2024).

Today, there are many playing styles, and the list keeps growing as new technologies like AI (Artificial Intelligence) arrive. With more data, analysts keep finding new styles, but as mentioned before, these are personal opinions rather than universal definitions. To help reduce the subjectivity of playing styles in football, a more scientific approach is needed. Here is where data-driven and machine learning approaches can be introduced to help us define tactical styles and how every team uses them in their playing styles. Using data-driven approaches can help us not only define tactical styles but also help us understand and improve them. It can also be useful for the coaching staff to better understand how their team is playing and what they can do to improve their team's performance. Also, a lot of football supporters and young coaches want to better understand how tactics work in football, and using data-driven approaches can help them to better understand different tactical styles and how they are played.

This research seeks to create a machine learning model that would associate tactical playing styles of every football team in the five major competitions (Premier League, Bundesliga, La Liga, Serie A, and Ligue 1) in the 2015/2016 season, and how similarly or uniquely teams played that season. This will contribute to helping reduce subjective opinions on defining tactical styles and rely more on data-driven and scientific approaches.

This work has been realized primarily with Python and a lot of other libraries such as sklearn, pandas, and socceraction for converting raw event data to more standardized data that will make it easier to work with, and a lot more. As said before, most of the code was generated with the help of generative AI tools and can be used openly by everyone on my GitHub account (See in the Notes for the link).

2. Literature review

2.1 The importance of playing styles in football

Recently, with the advancement of artificial intelligence, analysts and professionals have increasingly used machine learning tools to study and analyze football players and team performances based on large amounts of data. The goal is, of course, to help the manager and coaching staff make better decisions, such as choosing the right player for a specific position, correcting and developing individual players, making tactical adjustments, selecting the most suitable playing style, improving overall team performance, and much more.

Football is a simple sport where the team that scores more goals wins the game. It doesn't matter how they play, as long as they score more goals. Throughout history, and even in recent times, we've seen teams with fewer well-known players go on to win major trophies. One of the most famous examples is Greece at Euro 2004, a victory that many experts and fans saw as a miracle. But their win didn't happen by chance. Greece succeeded because it had a clear and effective tactical plan. They played a very defensive low block, staying compact and organized at the back, and combined it with quick counterattacks when they regained possession. This tactical style was criticized and blamed for not being very fancy or enjoyable to watch, but in the end, it is the success of the playing style that matters. Greece didn't have superstar names like Thierry Henry (France) or Luis Figo (Portugal), but what they did have was a well-organized team that worked together and followed their game plan from start to finish.

Greece's victory in 2004 is a great example of how important it is to choose the most suitable football playing style. A well-chosen tactical approach can be a key and important part of a team's success.

2.2 Coaches and football tactical styles

In today's football world, every team plays differently depending on the coach's tactical philosophy and the players available in the squad. It is important to understand that it is the coach who chooses a specific playing style, while the players on the pitch are responsible for executing the instructions given by the coach and his staff. At the amateur level, coaches often develop a playing style based on the available players. Since it is usually not possible to recruit specific players at this level, coaches must adapt their tactics to suit the strengths and limitations of their squad. On the other hand, professional football is a completely different story. At this level, coaches are often given a budget to recruit players who fit their tactical vision.

For example, when Pep Guardiola joined Manchester City in 2016, he transformed the team by signing players who matched his playing style. In his first season, he spent around 200 million euros and brought in seven new players. The following season, he spent approximately three hundred million euros and added six more players to the squad. In total, he signed 13 new players in his first two seasons. The reason behind these signings was that each player suited Guardiola's tactical philosophy. With this new squad, Guardiola began winning a lot of trophies in the seasons that followed (Orimolade [2020?]).

2.3 Characterizing tactical playing styles in football

Defining distinct playing styles in football has long been a subject of debate because, to this day, there is no clear way to categorize them into different groups. Many existing studies and articles on football "playing styles" reflect the subjective opinions of individuals and professionals, but there is still no universally accepted definition to categorize them. In recent years, with the development of new technologies, the collection of large amounts of data from football players through event data and tracking data, and the blooming advances in artificial intelligence, analysts are now better equipped to understand how different teams play. Today, almost every professional football club has at least one football analyst. Their job is to study player and team performances using large datasets and provide important information that the coaching staff can use to improve their team's performance on the pitch (AnalyiSport 2022).

2.3.1 The 5 key phases of a football game

In their study, proposed by (Hewitt, Greenham, Norton 2016), the authors clearly state that "there is not a definition for game style and consequently there has never been any attempt to measure game style" (Hewitt, Greenham, Norton 2016, p. 355). To address this issue, the author proposed to categorize football playing styles into 5 key phases, every phase has its own variables. These variables were selected based on high-level performance analysis and sports science literature. Many of these variables (or metrics) can be derived from event data like Wyscout or Statsbomb. Here is the following table, with the 5 key phases and the variables underneath:

Table 1: The five moments of play (source from (Hewitt, Greenham, Norton 2016))

Established Offense	**Transition from Attack to Defense**	**Established Defense**	**Transition from Defense to Attack**	**Set Pieces**
• Goal attempts	• Rate of change of	• Surface Area	• Ball speed	• Type of set piece

• Number of passes • Possession level • Passing sequences • Passing density • Passing efficiency and/or accuracy	centroid and surface area • Player density • Location of turnover • Player speed • Distance between players • Defensive transition speed	• Centroid position • Distance between players • Player density	• Rate of change of centroid and surface area • Player density • Number and length of passes • Player speed • Offensive transition speed	• Location versus success probability maps

This table is very helpful for my thesis, as it supports the idea that I can let the data reveal patterns by clustering variables defined in each phase of play, rather than subjectively predefining tactical styles. For example, in the phase “Established Offense,” variables like “possession level” and “goal attempts” can be used as features in a clustering model to identify distinct tactical patterns (Hewitt, Greenham, Norton 2016).

2.3.2 Exploratory Factor Analysis (EFA)

In the paper proposed by (Fernandez-Navarro et al. 2016), the authors proposed the following definition of what they considered to be a “style of play”: “A style of play could be considered as the general behavior of the whole team to achieve the attacking and defensive objectives in the game” (Fernandez-Navarro et al. 2016, p. 1). In this study, they found that a team’s style of play could be “defined by specific performance indicators and, consequently, teams can be classified to create a playing style profile” (Fernandez-Navarro et al. 2016, p. 1). The main method that they used was EFA (Exploratory Factor Analysis) to identify the key dimensions that define football playing styles among professional football teams in England and Spain. EFA is a very useful method when we want to group variables that behave similarly. Instead of having a large list of individual indicators, we can group them into more general indicators that they call “factors”. They first used 19 performance indicators (14 attacking indicators and 5 defensive indicators) in their study.

After collecting the 19 indicators, the authors then applied what we call PCA (Principal Component Analysis) as the first step of EFA. This method reduces the indicators into a smaller number of components. After that, they used the Varimax rotation, which is a technique that clarifies the factor structure for a better interpretation (Penn State's Department of Statistics [2024?]). By clustering correlated indicators, they found 6 football-style factors, which were the following:

Table 2: The 6 new factors identifying playing styles (source from (Fernandez-Navarro et al. 2016))

Factor	Name	Description	Style identified
1	Possession directness	Measures teams' progress with long direct passes or slowly built possession.	Direct vs. Possession style
2	Width of ball regain	Indicates where teams win the ball back	Wide regain vs. Central regain
3	Use of crosses	Measures whether teams rely on crosses in the final third	Crossing style vs. No-crossing
4	Possession width	How wide or narrow teams play when they have the ball	Wide vs. Narrow possession
5	Defensive pressure	How aggressively teams press without the ball	High pressure vs. Low pressure
6	Progression of attack	How fast teams go forward once they have the ball.	Fast progression vs. Slow progression

As we can see, these 6 factors define 12 different playing styles. In the study, the authors also put a score on each professional team in England and Spain for every factor, creating a playing style profile that characterizes how each team plays. This model or method shows us how complex tactical strategies in football can be reduced to a small set of meaningful dimensions.

2.3.3 Football analytical tools and metrics

To be able to discover football playing styles using supervised or unsupervised models, a variety of analytical tools and metrics have been developed and widely used in football analytics. These tools and metrics quantify different aspects of performance and style, evaluating every facet of player and team performance. Modern metrics capture a lot of details that can be used to better understand players and team performances (Gengee 2024).

2.4 Identifying tactical styles in football using machine learning

2.4.1 Spatio-temporal analysis using tracking data

A lot of work has already been done on how we can use machine learning to group different football teams into groups of tactical styles. The most notable one is the work done in the paper “Spatio-Temporal Analysis of Team Sports” by (Gudmundsson, Horton 2017), where the authors analyze the behavior of football players using tracking data. In their paper, they explain how modern tracking systems provide high-frequency spatio-temporal traces of player trajectories, which allow us to extract team performance insights based on movement and interactions rather than relying on metrics like possession or number of passes. Their entire study involves using tracking data to analyze the movement and positioning of football players. Another aspect is that they were focusing on the behavior of the players using heat maps and intensity matrices to show the presence of players in certain areas of a football pitch. The authors used an unsupervised clustering approach, which bears the name k-means. They didn’t predefine labels like “Center-forward” or “Wide-winger” but instead let the model do it by itself by clustering the players in different groups (Gudmundsson, Horton 2017).

2.4.2 Random forest and decision trees

In the article proposed by (Rein, Memmert 2016), the authors aimed to show how modern machine learning models like random forest and decision trees can be used to discover which tactical elements of play lead to successful attacks. By training on a large amount of historical match data, which was labeled with successful and unsuccessful outcomes, their models highlighted some key factors, for example, certain passing sequences significantly increase the chances of creating a goal-scoring opportunity. In the article, they discuss both classification and clustering models in the context of football tactical analysis. The authors highlight that supervised models (random forest, decision trees) are very useful when we have labeled data, such as match outcome (win, lose, or draw), passing sequence led to a goal, or player ratings based on performance. These models

are useful when our goal is to predict or explain an already known target label. They also talk about unsupervised models, in particular clustering, which are used when we don't have any labels, and we want to discover natural groupings of data in different categories. The unsupervised techniques that they mentioned are clustering, which detects team behaviors, formation changes, and role patterns. They also mentioned Bayesian latent models to identify group tactics and attack strategies by how similar they are (Rein, Memmert 2016).

2.4.3 Unsupervised clustering of tactical styles

Since defining playing styles is hard to do, using unsupervised clustering models can be a good way of finding different playing styles in football without predefining them in advance. Unsupervised methods have been widely used to discover tactical playing styles without prior labels, letting the data reveal the natural groupings of football playing styles. A common unsupervised technique is clustering, which we wrote about before, that is used when similar instances are grouped based on the features that we provided to the model. In one article, this approach was used to analyze how tracking data from football games can be segmented into multiple clusters to correspond to distinct playing styles (Gudmundsson, Horton 2017).

2.4.3.1 Topic modeling

An interesting method was developed using an unsupervised approach called the T3M (Team tactic topic model) to automatically discover tactical patterns at the professional football level. The model works on event data, focusing primarily on passing sequences from historical data. The idea is quite simple; this model is very similar to how topic models work in natural language processing, where they try to discover topics in a certain document, but here, instead, they try to discover tactical patterns in a sequence of passes. T3M learns to group these passing sequences into tactical patterns, each with a passing structure (a player that passes to another player) and a spatial structure (where the ball moves on the pitch). This allows us to discover team styles of play from raw data, like long passes to the forward might be a type of style widely common, which is called "Long ball" or short passing buildups on the wings might be another style of play, like "Possession based" or "Wing play" (Wang et al. 2015).

2.4.3.2 Graph-based

A graph-based unsupervised model was also studied to detect and compare football playing styles using a concept called "flow motifs" in pass networks. The authors define flow motifs as follows: "in which players are linked via executed passes, consist of a

given number of consecutive passes, namely, an ordered list of players who were involved in the particular passes" (Gyarmati, Kwak, Rodriguez 2014, p. 2).

The method works as follows:

1. A team's passing network is represented as a graph.
2. Players are represented as nodes in the graph.
3. Passes between players are represented as edges.
4. Motifs are small, repeating pass structures that capture the flow of the ball.

Instead of focusing on metrics like possession percentage or the total number of passes completed, this approach analyzes small pass sequences called motifs to reveal a team's tactical style. The study defined five types of 3-pass motifs (ABAB, ABAC, etc.) in advance. After building the pass network from a match, the researchers extracted all 3-pass sequences and converted them into one of the five motif types.

To understand whether certain motifs occurred more or less frequently than expected, they generated 1,000 randomized versions of each team's pass network and calculated z-scores (measures how much more or less a passing motif appears in a team's real matches compared to how often it would appear in a randomized version of those matches) for each motif type. At the end, they applied clustering methods such as k-means to group teams based on their motif usage patterns to organize teams by playing style (Gyarmati, Kwak, Rodriguez 2014).

2.4.3.3 A model proposing 19 playing styles

In their study (Plakias et al. 2023), the authors identified 19 different playing styles that they called factors. They used Exploratory Factor Analysis (EFA) on 88 tactical variables on 2999 matches to identify playing styles. Each factor. For example, factor 5 (Aerial game) included 4 metrics, such as the number of air challenges and the number of air challenges in the defensive third. Factor 2 (Transition game) had 7 metrics, such as the number of ball recoveries, the number of lost balls, and the number of wrong balls, among many more. This is a very important study; it used data-driven techniques and mathematics to discover different playing styles and not personal opinions (Plakias et al. 2023).

2.4.4 Supervised classification of tactical styles

Another way to discover tactical styles is to use a supervised classification method, which trains models on large sets of data using predefined labels. In this case, we already know

what the final target is, which is different from an unsupervised model, where the model would try to identify the target label by itself.

2.4.4.1 Bag of Words method

In their study (Montoliu et al. 2015), the authors developed a method to automatically recognize different types of football styles from video data. Their approach involved taking short video clips (about 10 seconds each) from football matches and converting them into feature representations. They predefined three target labels: ball possession, quick attack, and set piece. Each 10-second clip was labeled as one of these three categories. Several algorithms were used to train the model, including k-means, random forest, neural networks, and support vector machines. In the end, the best-performing algorithm was the random forest model, which achieved the highest results (Montoliu et al. 2015).

2.4.5 Limitations

Machine learning has been increasingly used in recent years for football analysis, with many studies applying machine learning methods to discover football styles (e.g., factor analysis with PCA, k-means clustering, sequential pattern mining, topic models, etc.). However, since machine learning is still a relatively new technology in the field of football analytics, existing methods are not yet advanced enough to provide highly precise or universally accepted categorizations of playing styles.

2.4.5.1 Limits of unsupervised clustering

Unsupervised methods are widely used by researchers to discover football playing styles without predefined target labels. However, their results heavily depend on the choice of input features and the clustering algorithms employed. Decisions such as which metrics to include or how many clusters to extract can lead to very different groupings of playing styles. Furthermore, the resulting clusters are more descriptive than predictive; researchers must interpret each cluster afterward, meaning that the definition of a football style is assigned subjectively. In other words, clustering helps identify patterns in the data, but labeling those patterns as specific tactical styles still requires subjective judgment. This subjectivity means that unsupervised style definitions can vary between studies.

2.4.5.2 Limits of supervised classification

Supervised classification approaches depend on labeled examples, which raises a fundamental problem: since football playing styles are not objectively defined, the training labels must come from experts and are inherently arbitrary. In practice, each coach or analyst might have their own opinion of what constitutes a particular style,

meaning there is no universally accepted target label. As a result, supervised approaches tend to inherit biases and subjective opinions, even if the best possible input features are selected for the model. If a time comes when a universal definition of playing styles is agreed upon, supervised models could become much more powerful and reliable for tactical analysis.

2.5 In summary

There were very few studies on discovering football playing styles using supervised models only. The main problem is that we don't have a clear definition of different playing styles, so we can't use target labels with 100% precision; most of the time, it is subjective opinions. In practice, the supervised approach often goes hand in hand and complements unsupervised approaches. Researchers might first opt to use unsupervised methods to discover football playing styles or patterns, and then, after that, they would label those patterns and train a supervised model to recognize them in new data.

Most of the studies related to my thesis were more concentrated on individual performances of football players. For example, some other studies were used to "classify elite and sub-elite goalkeepers in professional men's football" (Jamil et al. 2021, p. 1). A lot of studies were also used to study which player had the most goals in a season, most assists, most successful passes, most dribbles, most shots, most tackles, etc. These types of performances could be important for my thesis, but since I'm studying football playing styles, I'm more concentrated on patterns of play rather than individual performances. Of course, individual performances are important even in recognizing football styles, since there are key players in a football team that are crucial for a specific playing style, and without them, the team might suffer a lot.

2.6 Research direction

This thesis aims to create an unsupervised clustering model to classify or group teams into different tactical playing styles. We will use existing studies that will help us build our model by improving or adapting them. The features that we are going to use in our model are football analytics metrics identified by (Plakias et al. 2023) and modified and adapted by me. This study will rely on event data, as it is widely available through open-source platforms and offers sufficient volume and detail to compute most of the relevant tactical metrics that we need for our research.

3. Data types for football analytics

3.1 Event data

Event data, or what we can also call raw data, is the most common type of data available in football analytics. Though this type of data is mostly collected by specialists who watch football matches and record the events that happen on the pitch, nowadays the collection process is done semi-automatically, and the data is collected in real-time and is often available shortly after the game has ended. The main companies that provide this event data are StatsBomb, Opta, and Wyscout.

Event data includes a detailed description of every shot, pass, corner, dribble, foul, and much more. The data is very often stored in JSON or XML format, which makes it easier to work with. Event data is used for a wide range of purposes, mostly to describe what is happening in a football match. Here are some questions that event data tries to answer:

- How many shots did a player or a team make in a match?
- How many successful tackles did a player make?
- How many dribbles did a player attempt; how many were successful?
- How many fouls were committed in a match?
- How many key passes did a player or a team make in a match?
- Which player lost possession of the ball more often?

These are the types of questions that event data is used to answer. For many years, these questions were extremely valuable and helped provide more insight into football. For example, the first expected goals (xG) models were built using event data, helping teams make better decisions, such as taking fewer long shots (Oonk 2024a).

Even though event data is very useful and has many advantages, it also has its limits and disadvantages. The first limitation is that it may not be 100% accurate because of human error. Not all data is captured in every game, which can cause errors and gaps in the dataset, making analysis more difficult (Brandsen Sports 2023).

The second limitation is that event data sometimes lacks context. For example, it doesn't show whether a shot was taken against an empty goal or a crowded defense. This is why, when we need more detailed context, we use tracking data instead.

StatsBomb provides match datasets with 72 rows, excluding the match ID. Thus, we have 72 types of information available for a football match. In the first 10 rows, for

example, we have the minute of the match, which is an integer value; the possession of the match, which is also an integer value; and the type_name, which tells us if there was a pass, shot, carry, ball recovery, and more.

3.2 Tracking data

Tracking data records the position of all players and the ball at a frequency usually between 10 and 25 times per second. The most common way to collect this data is through cameras. The video footage collected throughout the game is analyzed by deep learning models that track the players and the ball. Other technologies like LPM, GPS, or radio-based devices can also be used.

Tracking data is mainly collected in stadiums using cameras and machine learning. However, at the start of each match, someone must manually assign players' names to their respective positions on the field, because the model is able to see the location of players but not their identities. Sometimes, during celebrations, when players group, the model loses track of who is who. In those cases, manual corrections are needed. As mentioned earlier, event data lacks context, and tracking data fills that gap. For example, with event data, we know when a certain player makes a pass or a shot, but we don't know what the other players on the pitch are doing at the same time. With tracking data, however, we can see exactly the position and actions of every other player on the pitch at that moment.

At first, tracking data was mainly used to analyze players' physical performance, such as the distance covered, speed, and number of sprints. Nowadays, its use has expanded a lot. For example, tracking data can help with the following problems:

- Analyze pitch control (whoever controls different areas of the field)
- Improve expected goals (xG) models by adding context
- Assess passing options (which players are available to receive the ball)
- Evaluate pressing effectiveness (how players react under pressure)
- Study the team’s defensive structures.

The biggest advantage of tracking data is that it adds important context to match events.

The disadvantages of tracking data are, first, the very limited availability of open-source or free datasets on the internet. Most open tracking datasets only cover a few matches, and accessing comprehensive tracking data is expensive, which restricts researchers’ ability to use them. Another issue is that signal errors or interpretation mistakes can occur during a match, which can lead to inaccuracies and become problematic when using the data for research purposes (Oonk 2024b).

4. Methodology

Several modeling approaches to classify teams into tactical styles were introduced, which were classification and clustering. Of these two, based on my findings and analysis, the clustering approach was more suitable. By using clustering models, we are not predefining tactical styles but letting the model do so, which is very important because that takes away the subjective approach and favors the objective approach.

4.1 Data source used

In this project, we will be using open-source event data from Statsbomb, which we can download through GitHub or use their library, which we can install and import into our project. Due to conversion difficulties when using Valuing Actions by Estimating Probabilities (VAEP) with the socceraction library (KU Leuven 2025; 2020), I downloaded the whole Statsbomb open-source data from GitHub (Statsbomb 2025). Event data, as described before, contains a detailed description of every shot, pass, corner, dribble, foul, and much more in a single football match. Some events also have contextual information like timestamps, coordinates on the pitch, player ID, and the teams that they are playing in. The study focuses on various competitions; here are the 5 competitions that we are using to find the best model to classify teams into tactical styles:

- Premier League 2015/2016 season
- Bundesliga 2015/2016 season
- Ligue 1 2015/2016 season
- La Liga 2015/2016 season
- Serie A 2015/2016 season

The goal is to analyze and compare different competitions and see how teams group together based on tactical styles, which is why we have so many competitions in different countries. We can notice that we are choosing the same season, which is the 2015/2016 season. There is one main reason for that: the 2015/2016 season is one of the few in the Statsbomb open dataset that includes all matches, so we can build our model based on a full season, which is important.

4.2 Data structure

Statsbomb provides structured event information in JSON format. The key components are the following:

Competitions.json

Figure 1: Example of what the Competitions.json file contains

	competition_name	season_name	competition_id	season_id
0	1. Bundesliga	2023/2024	9	281
1	1. Bundesliga	2015/2016	9	27
2	African Cup of Nations	2023	1267	107
3	Champions League	2018/2019	16	4
4	Champions League	2017/2018	16	1
..	...	...	...	...
69	UEFA Euro	2020	55	43
70	UEFA Europa League	1988/1989	35	75
71	UEFA Women's Euro	2022	53	106
72	Women's World Cup	2023	72	107
73	Women's World Cup	2019	72	30

This JSON file contains a list of all available competitions and seasons as IDs. For example, we can look for a competition for a specific season, like if we use the competition_id number 9 and season_id number 27, we get the event data of the Bundesliga 1 2015/2016 season.

Matches

Figure 2: Example event data for a specific season

	game_id	season_id	competition_id	competition_stage	game_day	\
0	3890561	27	9	Regular Season	34	
1	3890505	27	9	Regular Season	28	
2	3890511	27	9	Regular Season	29	
3	3890515	27	9	Regular Season	29	
4	3890411	27	9	Regular Season	17	
..	...	...	...	...	...	
301	3890263	27	9	Regular Season	1	
302	3890262	27	9	Regular Season	1	
303	3890261	27	9	Regular Season	1	
304	3890260	27	9	Regular Season	1	
305	3890259	27	9	Regular Season	1	

	game_date	home_team_id	away_team_id	home_score	away_score	\
0	2016-05-14 15:30:00	175	181	1	4	
1	2016-04-02 15:30:00	169	184	1	0	
2	2016-04-08 20:30:00	173	178	2	2	
3	2016-04-09 15:30:00	171	872	1	2	
4	2015-12-20 16:30:00	173	177	2	0	
..	...	...	...	...	...	
301	2015-08-15 15:30:00	177	189	0	1	
302	2015-08-15 15:30:00	872	178	2	2	
303	2015-08-15 18:30:00	180	185	4	0	
304	2015-08-15 15:30:00	904	175	2	1	
305	2015-08-14 20:30:00	169	171	5	0	

	venue	referee
0	PreZero Arena	Felix Brych
1	Allianz Arena	Florian Meyer
2	Olympiastadion Berlin	Benjamin Brand
3	Volksparkstadion	Peter Sippel
4	Olympiastadion Berlin	Peter Sippel
..	...	...
301	Mewa Arena	Guido Winkmann
302	Merck-Stadion am Böllenfalltor	Felix Brych
303	Signal-Iduna-Park	Tobias Stieler
304	BayArena	Robert Hartmann
305	Allianz Arena	Bastian Dankert

This is a directory containing a JSON file for a specific season; each file contains information about the competition and season. We can see an example above of the Bundesliga 1 2015/2016 season and all the information. The JSON file is found in the following directory: matches/9/27.json.

Events

Figure 3: Example of event data for a specific match

	game_id	event_id	period_id	team_id	\
0	3890481	24584357-1d61-44ae-bc25-b2477d93911f	1	180	
1	3890481	236c857d-58b6-4bb2-91ff-cba1eae824d3	1	169	
2	3890481	b28be44c-6f07-4829-86ca-fc28b9563366	1	169	
3	3890481	8ce97a84-c220-40cb-8a6c-73278e20168b	1	180	
4	3890481	7da63b6f-f6e6-4c7d-9d08-f0fe4ab63c6f	1	180	
...	...	...	...	...	
4012	3890481	68eec2ec-1c7b-43f7-9e02-77420e6de37f	2	180	
4013	3890481	547a90fd-7dc5-48fc-9c38-865578d1dba0	2	180	
4014	3890481	a8d7d269-4a3c-401d-b49f-dde02b31175a	2	180	
4015	3890481	9a320728-3f6f-4261-9e2c-48c645389d80	2	180	
4016	3890481	0bf646ca-0b49-4a11-a2ac-af2a8764a21e	2	169	

	player_id	type_id	type_name	index	timestamp	minute	\
0	NaN	35	Starting XI	1	0 days 00:00:00	0	
1	NaN	35	Starting XI	2	0 days 00:00:00	0	
2	NaN	18	Half Start	3	0 days 00:00:00	0	
3	NaN	18	Half Start	4	0 days 00:00:00	0	
4	3712.0	30	Pass	5	0 days 00:00:00.981000	0	
...	...	...	...	...	...	...	
4012	8225.0	42	Ball Receipt*	4013	0 days 00:46:04.805000	91	
4013	8225.0	43	Carry	4014	0 days 00:46:04.805000	91	
4014	8225.0	30	Pass	4015	0 days 00:46:05.991000	91	
4015	NaN	34	Half End	4016	0 days 00:46:07.104000	91	
4016	NaN	34	Half End	4017	0 days 00:46:07.104000	91	

	...	team_name	duration	\
0	...	Borussia Dortmund	0.000000	
1	...	Bayern Munich	0.000000	
2	...	Bayern Munich	0.000000	
3	...	Borussia Dortmund	0.000000	
4	...	Borussia Dortmund	1.153855	
...	...	...	...	
4012	...	Borussia Dortmund	NaN	
4013	...	Borussia Dortmund	1.185701	
4014	...	Borussia Dortmund	1.242911	
4015	...	Borussia Dortmund	0.000000	
4016	...	Bayern Munich	0.000000	

This is a directory containing a specific match; it contains all on-ball actions of a particular match. Above, we can see information from Borussia Dortmund vs Bayern Munich. There are a lot more events available, but since there are so many, I'm only showing parts of them above. The JSON file is found in the following directory: events/ 3890481.json.

4.3 Selected and designed metrics

4.3.1 Transforming event data into SPADL format

While raw event data from Statsbomb are in a very rich format, these raw data are not very suitable for modeling sequences of actions in a structured and consistent way. Here

is where we need to use what we call Soccer Player Action Description Language (SPADL) (KU Leuven 2020). SPADL was developed as a part of the socceraction Python library to provide a standardized and clear representation of in-game actions. It was designed mainly to value player actions, which also gives us a broader analysis of how the team plays. Each action is converted into a set of fields with standardized units and names. I came across SPADL when I was working with VAEP metrics, and since then, because of how useful it is, I have used it in all my coding, even though I was not using VAEP metrics a lot. All the metrics that I use in my study derive from football actions using SPADL.

4.3.2 VAEP metrics

My first model had only VAEP metrics. Here is the list of metrics that I used:

1. VAEP_Passes
2. VAEP_Crosses
3. VAEP_Dribbles
4. VAEP_Shot
5. VAEP_Clearance
7. VAEP_Interception
8. VAEP_Foul
9. VAEP_Tackle
10. VAEP_BadTouch
11. VAEP_KeeperSave

There are 11 different metrics. I chose these metrics based on the study done by (Decroos et al. 2019), where they introduced a new framework that would estimate how much an action would change a team's probability of scoring or conceding a goal. At the end of the article, they showed all the action types that were used in their framework, and I took some of these action types and transformed them into metrics (Decroos et al. 2019). So, I mainly put the metrics that I thought would describe different playing styles. For example, teams with a high score of VAEP_Passes are the teams that would be more likely to score goals based on the sequence of passes, which also means that possession-based teams would have a high score in VAEP_Passes.

The problem with only using VAEP metrics is that we are only using metrics based on actions; this is what VAEP is about, it only gives a score based on actions. So, what it will do is that if we take VAEP_Passes it will give a score based on the probability of conceding or scoring a goal after the pass is made. It does sound good in the beginning, but the problem is that the teams that pass the ball a lot will always be in favor of having

a higher VAEP score, no matter if they lose the ball more often than other teams. So, we can see the limits of relying only on VAEP metrics; the team that has the ball the most has higher VAEP scores in most of the metrics, which makes the model have bad results and cluster teams into styles based on VAEP performances rather than tactical styles.

4.3.3 From SPADL to tactical features

For some metrics, particularly those derived from VAEP, I used an XGBoost model to compute the score to make these metrics perform better. What I did was train an XGBoost model on the 5 competitions that we are analyzing. What this model does is take game after game and calculate how much each action in a match contributes to scoring or conceding a goal in the next 10 actions. For example, if a midfielder has the ball in the midfield and passes it to a striker who is very close to the goal, based on the past data, VAEP will give a positive or negative score depending on whether the pass that the midfielder made to the striker would lead to a goal or concede a goal. The more we train the XGBoost model, the better the metrics that use VAEP scores will perform.

In the very beginning, I did not train an XGBoost model for some metrics that required VAEP scores. I would simply calculate the VAEP match per match without pretraining. This was not optimal because my VAEP model did not have much historical data, which makes it unreliable.

After that, what I did was to pretrain an XGBoost model on one season only. For example, when I was analyzing the Premier League 2015/2016 season, I had pretrained an XGBoost model for all 380 matches of that season to calculate the VAEP, and I would use the score for all the metrics that require VAEP. It was much better because I have a VAEP model that is well-trained and will perform much better on metrics that require VAEP scores.

Finally, I concluded that if I wanted to make the XGBoost model more effective and reliable and be useful for all 5 competitions combined, I had to pretrain the VAEP XGBoost model on all 5 competitions and use it for every competition.

4.3.4 Other metrics beyond VAEP

In this model, I used VAEP metrics and added other well-known metrics that don’t use VAEP. These are the following metrics that I added:

1. PPDA
2. Verticality
3. ProgressivePasses
4. Directness/ BuildUpSpeed

Now let's look into these metrics. PPDA is a well-known defensive metric that is used to measure the pressing intensity of a team; the lower the score of PPDA, the more a team presses. PPDA is calculated as the number of passes made by the opposite team divided by the number of defensive actions made by the team that does not have the ball. Defensive actions are, for example, tackles, interceptions, fouls, or challenges (Tweedale 2023).

Directness, Verticality, and ProgressivePasses are tempo-based metrics, which will measure the speed and rhythm at which a team progresses in possession-based phases. Verticality measures how quickly a team moves with the ball towards the opponent's goal; it is calculated as meters gained per second. Directness measures the "ratio of forward passes to lateral or backward passes" (The Football Analyst 2025a). The higher the score, the more the team plays forward, and the lower the score, the more the team plays horizontally and in the back (The Football Analyst 2025a). ProgressivePasses is the speed and urgency with which a team must go forward, measuring passes that bring the ball closer to the opposition's goal. For example, a defender who makes a 15-meter pass forward counts as a progressive pass because we are moving the ball significantly towards the opponent's goal, and the attacking phase is advancing (The Football Analyst 2025b).

These are more tempo and rhythm-based metrics and go more into the behavior of a team and players on the pitch. I feel that this is important because football is not only about passes and goals but also about human emotions and behavior, which are harder to calculate.

4.3.5 EFA-related indicators

These 19 indicators that I used in my model were proposed by (Fernandez-Navarro et al. 2016) in their study (see section 2.3.2). They used 19 performance indicators, 14 attacking and 5 defensive indicators, and at the end, using PCA and Varimax rotation, they found 6 different playing styles. I used their 19 indicators as metrics to see whether they can be useful to classify teams into tactical styles.

4.3.6 The 19 playing styles adapted

This is my final model, which also classifies the teams the best. What I did is that I used the metrics that (Plakias et al. 2023) found in their study, using more than 88 performance indicators in nearly 3000 matches across 11 European leagues, and they identified 19 factors explaining more than 80% of the variance in playing styles. I tried to use almost all of them, but some I couldn't use, like the factors 18 and 19, because event data doesn't have offsides, so I could not use them or make an approximation of offsides.

Some of the metrics that I used are more of an approximation because of a lack of event data or, more importantly, a lack of tracking data. For example, factors 18 and 19, which are offside, need tracing data, as well as some others like factor 1, which is elaboration of buildup, and factor 10, which is pressing. Here is the list of metrics that I used:

1. FinalAttemptCreation
2. EffectiveTime
3. BuildUpSpeed
4. LowBlockRatio
5. Fouls
6. TimePerAction
7. Recoveries_OwnHalf
8. AerialChallenges
9. Dribbles
10. DefensiveActions
11. PPDA
12. OppSetPieceRatio
13. OppOpenPlayRatio
14. AvgDefensiveRegainX
15. YellowCards
16. LostBalls
17. SetPieceAttackRate
18. PreventedCounters_Mid3
19. WingPlayIndex
20. CentralPlayIndex
21. Crosses
22. CounterattackIndex
23. OppCounterAttackRate

We can see that we have a total of 23 metrics. For some factors, I could find multiple metrics. For example, factors 2, 8, 12, 1, 4, and 17 have at least 2 metrics, others have 1. These are the metrics that will be used to find the best model that clusters teams into tactical styles. If I associate these metrics with every factor proposed by (Plakias et al. 2023). We have the following table that illustrates that:

Table 3: Attributing metrics for every style

Number	Factor name	Metrics used

1	Elaboration of Build-Up Phase	BuildUpSpeed
2	Transition Game	Recoveries_OwnHalf, LostBalls
3	Attacking Transition	CounterattackIndex
4	Defensive Transition	OppCounterAttackRate, PreventedCounters_Mid3
5	Aerial Game	AerialChallenges
6	Type of Attack	SetPieceAttackRate
7	Crossing	Crosses
8	Type Of Opponent Attack	OppOpenPlayRatio, OppSetPieceRatio
9	Defensive Blocks	LowBlockRatio, AvgDefensiveRegainX
10	Press	PPDA
11	Individual Defensive Actions	DefensiveActions
12	Width of Creative Phase	WingPlayIndex, CentralPlayIndex
13	Effective Game	EffectiveTime
14	Individual Attacking Actions	Dribbles
15	Tendency To Create Final Attempts	FinalAttemptCreation
16	Passing Tempo	TimePerAction
17	Defending Aggressively	Fouls, YellowCards

4.4 Tactical feature selection and model construction

The hardest part of this study was choosing the best metrics/features for my model so it can classify football teams into similar tactical styles. I tried a lot of different types of metrics, but they were not clustering the teams in the correct way or in the correct style of play. The model using only VAEP metrics, as we said before, favors teams that have more possession of the ball, which is not what we want; we want to identify how teams

played tactically, not how good their performances were. The model that blends traditional metrics and VAEP metrics has better results, but we are still using a lot of VAEP metrics, which favors teams with possessions. The model that uses 19 EFA-related indicators gives us very bad results compared to all the models that I tried; it clustered teams into the same tactical styles even though it was not the case that season, and it failed to capture how uniquely Leicester City played tactically that season. The main reason that these models were not good is that it was hard to associate teams with tactical styles.

Here is where the model proposed by (PLAKIAS et al. 2023) comes in and takes the lead, the table that they proposed and that I modified (see section 4.3.6) tells us what style of play is associated with every metric, so when a team or group of teams has a high value in a specific metric, we can directly identify the playing style based on the table. For example, Leicester City had the lowest score in PPDA of all the teams in the Premier League. If we look at the table, PPDA is associated with the playing style “Press”, so one of the playing styles of Leicester City was that they pressed a lot that season.

4.5 Clustering techniques

For each model per competition, I applied several unsupervised clustering algorithms to see which one classifies teams most optimally into similar tactical styles, which is our goal in this study. I used a total of 5 different algorithms per model. The goal was to use as many clustering algorithms as possible and to be able to choose the one that performs the best, based on 3 evaluation metrics that we will see shortly. Here are the following clustering algorithms used in my study:

1. K-Means clustering

2. Agglomerative Hierarchical clustering

3. Gaussian Mixture Models clustering (GMM)

4. Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH)

5. Spectral clustering

Every clustering algorithm has its pros and cons, but our goal is to see which one classifies teams into tactical style the best, so we care more about what the results are rather than the advantages or disadvantages each clustering algorithm has.

4.5.1 Evaluation metrics

As mentioned before, for every clustering algorithm, I needed to choose a predefined number of clusters, but how can I know how many numbers of clusters each algorithm needs to perform the best? To solve that problem, I used 3 types of clustering evaluation metrics:

1. Silhouette Score

2. Calinski-Harabasz Index

3. Davies–Bouldin index

The Silhouette score measures how well each data point (in my case, football teams) is similar to its clusters compared to other clusters. It considers two aspects. 1) Cohesion: “how close a data point is to other points in its own cluster” (Farshad K 2024). 2) Separation: “How far a data point is from the points in the nearest neighboring cluster” (Farshad K 2024). The scores go from -1 to +1; the closer we are to +1, the more the data point is matched to its cluster and far from other clusters, indicating more compact and better separated clusters (Farshad K 2024).

The Calinski-Harabasz Index looks at the bigger picture; it calculates how far clusters are from each other (between-cluster dispersion) to how close points are to each other (within-cluster dispersion). The higher the score, the better the clustering, because clusters are “closer together (more dense), while clusters themselves are further away from each other (well separated)” (Misha Sv 2022).

The Davies–Bouldin index evaluates the quality of the clusters by how similar each cluster is to another similar cluster, and then it takes these pairs of clusters that are similar and gives them a score on how well separated they are and how compact they are. The higher the score, the worse the clustering performance is. The lower the score, the better. “It indicates that the clusters are well-separated and compact” (GeeksforGeeks 2025).

4.5.2 Number of cluster configurations

Every model in every competition had different scores, so evaluation metrics were used in every model to see what the optimal number of clusters was. But there was one problem when I was working on the global model; here we had to define the same number of clusters for every model because, as I said before, I wanted a general model that would work on every competition. Let’s see how these evaluation metrics score the models for the Premier League, the optimal number of clusters based on the evaluation

metrics was 9 for k-means and 7 for GMM, Agglomerative and BIRCH but for Spectral it was 3 clusters, the first 4 models are alright in terms of numbers but for Spectral it did not make sense to have that little amount of clusters if the goal is to cluster teams into different tactical styles, because with only 3 clusters we don't get many different tactical styles, we only get 3 major tactical styles, and the teams that played a unique tactical styles will not be clustered alone, this is why for every model that suggest lower number of clusters like the one with Spectral but not only, most of the models that proposed less than 6 clusters I would predefine 6 numbers of clusters for them even though if the performance would be slightly worse, I preferred more diverse tactical styles. Here, underneath, we have an example of how many clusters it was optimal for a Premier League model:

Figure 4: Premier League evaluation metrics per clustering algorithm

KMeans

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.178	3.96	1.816
3	0.176	4.21	1.583
4	0.169	4.08	1.371
5	0.194	4.36	1.181
6	0.179	4.27	0.945
7	0.170	4.46	0.826
8	0.173	4.03	0.944
9	0.205	4.83	0.695

Agglomerative

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.126	3.79	2.130
3	0.152	4.09	1.598
4	0.179	4.36	1.440
5	0.189	4.44	1.190
6	0.201	4.49	1.058
7	0.221	4.53	0.956
8	0.217	4.71	0.834
9	0.196	4.75	0.924

GMM

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.123	3.63	2.135
3	0.099	3.05	1.537
4	0.104	3.24	1.347
5	0.099	3.12	1.255
6	0.114	3.41	1.083
7	0.187	3.99	0.948
8	0.175	3.97	0.879
9	0.185	4.22	0.792

Birch

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.126	3.79	2.130
3	0.152	4.09	1.598
4	0.179	4.36	1.440
5	0.189	4.44	1.190
6	0.201	4.49	1.058
7	0.221	4.53	0.956
8	0.217	4.71	0.834
9	0.196	4.75	0.924

Spectral

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.140	3.87	2.044
3	0.165	3.77	1.651
4	0.139	3.90	1.496
5	0.098	3.72	1.392
6	0.141	4.05	1.033
7	0.007	2.92	1.565
8	-0.075	1.97	2.010
9	-0.102	2.33	1.542

As we can see, the optimal was 9 clusters for k-means and 7 for GMM, Agglomerative, and BIRCH, but for Spectral it was 3. As I said, I predefined 6 clustering algorithms in every model where the number of clusters was suggested to be lower than 6. Here we have another example with the Bundesliga:

Figure 5: Bundesliga evaluation metrics per clustering algorithm

KMeans

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.332	6.22	1.180
3	0.302	7.18	1.133
4	0.364	11.09	0.825
5	0.303	10.28	0.672
6	0.293	11.63	0.698
7	0.304	12.80	0.602
8	0.273	13.73	0.499
9	0.281	16.78	0.498

Agglomerative

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.332	6.22	1.180
3	0.345	7.18	0.839
4	0.351	8.48	0.681
5	0.351	9.19	0.567
6	0.293	11.63	0.698
7	0.304	12.80	0.602
8	0.308	13.80	0.588
9	0.281	16.78	0.498

GMM

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.347	6.04	1.443
3	0.260	6.37	1.010
4	0.211	6.05	1.088
5	0.282	10.27	0.809
6	0.216	8.95	0.788
7	0.085	8.87	0.753
8	0.235	12.34	0.585
9	0.209	14.34	0.495

Birch

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.332	6.22	1.180
3	0.345	7.55	0.972
4	0.357	8.82	0.765
5	0.364	9.62	0.645
6	0.234	8.88	0.542
7	0.196	8.25	0.423
8	0.253	12.78	0.458
9	0.236	13.26	0.360

Spectral

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.266	5.18	1.547
3	0.311	6.49	1.025
4	0.360	11.00	0.825
5	0.295	10.52	0.841
6	0.111	7.54	1.071
7	-0.031	5.30	1.553
8	-0.106	4.94	2.046
9	-0.074	4.61	1.219

We can see that the scores for the Bundesliga model are quite different from the Premier League model. The preferred number of clusters based on the evaluation metrics is much lower than in the Premier League, and they also have larger scores in general. This is because there are fewer teams in the Bundesliga, which alters the scores of the evaluation metrics. But as I said before, I will not go down 6 clusters for every model because I need a wider range of tactical styles.

4.6 Choosing the best models per competition

To choose the best model per competition and then also the best global model across all competitions is very challenging; the easiest way is to compare how relevant these models and their clusters are, based on external sources. But for my study, I wanted and I was trying to propose my solutions based only on a mathematical and evaluation-based way, so the goal was not to use external sources to see if my models are correct in

grouping teams into similar or unique tactical styles, but rather to use what I found using evaluation metrics like the ones that we saw above (Silhouette Score, Calinski-Harabasz Index, and Davies–Bouldin index) and also use the 23 metrics that I proposed (see section 4.3.6). In the following chapters, I will show you 2 stages that I used to choose the best models.

4.6.1 Stage 1

This is the first step to choosing the best model by using evaluation metrics. I would compare all the scores of the different clustering algorithms and then, at the end, choose the clustering algorithm that had the best scores. Before that, we need to understand what the scores mean in each of the evaluation metrics.

1. Silhouette Score: the higher, the better, mostly between 0 and 1, but can also go negatively from 0 to -1

2. Calinski-Harabasz Index: the higher, the better, between 0 and infinity

3. Davies–Bouldin index: the lower, the better, between 0 and infinity

Also, as I said before, we want to use more than 6 clusters, so everything underneath would not be looked at, even though they might be better scored. Here, I prepared a table that explains every step in stage 1:

Table 4: Steps for stage 1

Step	Description	Outcome
1	Examine the overall score of all 5 clustering algorithms and identify any anomalies.	Discard algorithms with poor performance overall compared to other algorithms.
2	For each clustering algorithm, we need to identify the best number of clusters based on the 3 evaluation metrics. The preferred number of clusters is based on these criteria: - Highest score in 3 evaluation metrics. If that is not possible, go to the next criterion. - Highest score in two out of three evaluation metrics. If that is not possible, go to the next criterion.	Determines the optimal number of clusters for every clustering algorithm. We will have 5 models, one for every clustering algorithm.

	- Highest score in Silhouette score.	
3	In this step, we rank the final 5 selected algorithms by using the same step as we did in step 2, where we used different criteria.	Final ranking of clustering algorithms from best to worst in every competition.

Now that we know these steps, here is an example of how I choose the best model in the Premier League using the 3 steps:

1) Step 1, I would look at the scores in general of all 5 clustering algorithms to see if there are any anomalies at first sight (see Figure 4, section 4.5.2). For the Premier League, we can see that Spectral has some negative values, and the values in general for all the evaluation metrics are poorer, so I would discard that. Sometimes this might not be the case for other competitions, but in this case, we can see some anomalies.

2) Step 2 is to analyze every algorithm individually to see how many clusters are the best based on the scores. For example, with k-means, the last one with 9 clusters performs the best:

Figure 6: Evaluation metrics for k-means for the Premier League

KMeans

Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
2	0.178	3.96	1.816
3	0.176	4.21	1.583
4	0.169	4.08	1.371
5	0.194	4.36	1.181
6	0.179	4.27	0.945
7	0.170	4.46	0.826
8	0.173	4.03	0.944
9	0.205	4.83	0.695

It has the highest score in Silhouette, Calinski-Harabasz, and the lowest in Davies–Bouldin, which is by far the best. Sometimes it is harder because not all of them might have the best score. In that case, we will look for at least 2 of the 3 evaluation metrics to have the highest score. If that is still not possible, then we will base our logic on which one has the highest Silhouette score.

At the end, we will have the optimal number of clusters for every clustering algorithm. Here are examples of the 5 best in the Premier League:

- K-means: 9 clusters, where all three evaluation metrics performed the best

- Agglomerative: 8 clusters, it has the lowest score in the Davies–Bouldin metric, and compared to cluster 9, it has a better silhouette score.
- GMM: 9 clusters, has the best score in 2/3 metrics
- BIRCH: 8 clusters. This algorithm has the same values as the Agglomerative algorithm, so it will produce the same clusters. BIRCH and Agglomerative sometimes have the same scores
- Spectral: Discarded because it had anomalies in the beginning

At the end, we have these models and their scores:

Figure 7: Final ranking of the best models per clustering algorithm

Name	Clusters	Silhouette	Calinski-Harabasz	Davies-Bouldin
KMeans	9	0.205	4.83	0.695
Agglomerative	8	0.217	4.72	0.834
GMM	9	0.185	4.23	0.792
Birch	8	0.217	4.72	0.834

3) Step 3, the final step, is to rank them from the best to the worst based on the scores, like we did before:

   K-means is the one that has the best scores in 2/3 metrics, so it is the first. Next are Agglomerative and BIRCH, both in 2nd place and 3rd place; it is GMM that ranks the poorest among the other algorithms. Finally, for the Premier League, the best models are the following:

   1st K-means 9 clusters.

   2nd Agglomerative and BIRCH 8 clusters.

   3rd BIRCH 8 clusters.

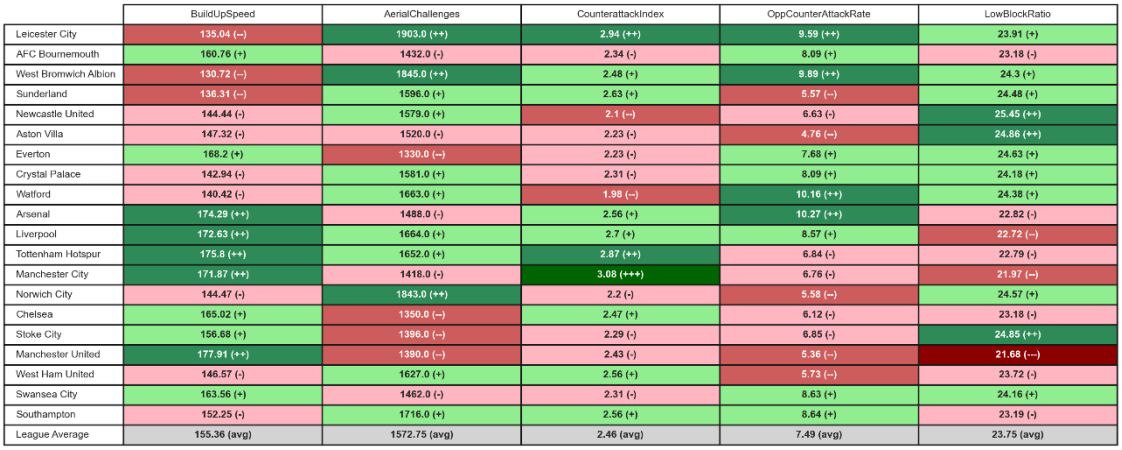
Now, what we will do is move to stage 2 to find tactical styles for every cluster.

4.6.2 Stage 2

After stage 1 is completed, we can move to stage 2. We will take the best model that was distinguished in stage 1 and analyze it. I will apply to each cluster its corresponding factors with the help of the 23 metrics and the teams that were in that cluster.

Below we can see an example of the values of 5 out of 23 metrics per team in the Premier League (see Appendix 1 for the full figure):

Figure 8: Scores of every team per metric.



	BuildUpSpeed	AerialChallenges	CounterattackIndex	OppCounterAttackRate	LowBlockRatio
Leicester City	135.04 (--)	1903.0 (++)	2.94 (++)	9.59 (++)	23.91 (+)
AFC Bournemouth	160.76 (+)	1432.0 (-)	2.34 (-)	8.09 (+)	23.18 (-)
West Bromwich Albion	130.72 (--)	1845.0 (++)	2.48 (+)	9.89 (++)	24.3 (+)
Sunderland	136.31 (--)	1596.0 (+)	2.63 (+)	5.57 (--)	24.48 (+)
Newcastle United	144.44 (-)	1579.0 (+)	2.1 (--)	6.63 (-)	25.45 (++)
Aston Villa	147.32 (-)	1520.0 (-)	2.23 (-)	4.76 (--)	24.86 (++)
Everton	168.2 (+)	1330.0 (--)	2.23 (-)	7.68 (+)	24.63 (+)
Crystal Palace	142.94 (-)	1581.0 (+)	2.31 (-)	8.09 (+)	24.18 (+)
Watford	140.42 (-)	1663.0 (+)	1.98 (--)	10.16 (++)	24.38 (+)
Arsenal	174.29 (++)	1488.0 (-)	2.56 (+)	10.27 (++)	22.82 (-)
Liverpool	172.63 (++)	1664.0 (+)	2.7 (+)	8.57 (+)	22.72 (--)
Tottenham Hotspur	175.8 (++)	1652.0 (+)	2.87 (++)	6.84 (-)	22.79 (-)
Manchester City	171.87 (++)	1418.0 (-)	3.08 (+++)	6.76 (-)	21.97 (--)
Norwich City	144.47 (-)	1843.0 (++)	2.2 (-)	5.58 (--)	24.57 (+)
Chelsea	165.02 (+)	1350.0 (--)	2.47 (+)	6.12 (-)	23.18 (-)
Stoke City	156.68 (+)	1396.0 (--)	2.29 (-)	6.85 (-)	24.85 (++)
Manchester United	177.91 (++)	1390.0 (--)	2.43 (-)	5.36 (--)	21.68 (---)
West Ham United	146.57 (-)	1627.0 (+)	2.56 (+)	5.73 (--)	23.72 (-)
Swansea City	163.56 (+)	1462.0 (-)	2.31 (-)	8.63 (+)	24.16 (+)
Southampton	152.25 (-)	1716.0 (+)	2.56 (+)	8.64 (+)	23.19 (-)
League Average	155.36 (avg)	1572.75 (avg)	2.46 (avg)	7.49 (avg)	23.75 (avg)

As we can see, we have the score for every team and the league average in the last column of the table. There are also some signs near the scores like -, --, ---, and +, ++, +++. Here is a table explaining what the signs mean:

Table 5: Explanation of the colors and the signs

Sign (with associated color)	Meaning
-	The score of the metric is below the average score.
--	The score of the metric is below the average score. Here we use one standard deviation below the average.
---	The score of the metric is way below the average score. Here we use 2 standard deviations below the average.
+	The score of the metric is above the average score.
++	The score of the metric is above the average score. Here we use 1 standard deviation above the average.

+++	The score of the metric is way above the average score. Here we use 2 standard deviations above the average.

Knowing this, when we cluster our models using all the different clustering algorithms, we can more easily find tactical patterns and names of the tactical system of a specific cluster. For example, we can see that Leicester City, Manchester City, and Tottenham Hotspur are the only teams that have a ++ in CounterattackIndex. We can say that one of their tactical styles/behaviors is playing counterattacks.

We also need the 17-factor table proposed by (Plakias et al. 2023) that I modified and its corresponding metrics (see section 4.3.6).

Now that we have the 2 tables and the clustering algorithm for the Premier League (k-means with 9 clusters determined in stage 1), let's see how we can associate tactical styles for every cluster:

K-means with 9 clusters had the following teams per cluster:

Figure 9: K-means clustering for the Premier League

```
KMeans Clustering
Silhouette Score: 0.205
Calinski-Harabasz: 4.83
Davies-Bouldin: 0.695
Cluster composition:
   Cluster 0: ['AFC Bournemouth', 'Everton', 'Arsenal', 'Stoke City', 'Swansea City']
   Cluster 1: ['West Bromwich Albion', 'Crystal Palace', 'Norwich City', 'Southampton']
   Cluster 2: ['Liverpool', 'Tottenham Hotspur']
   Cluster 3: ['Sunderland']
   Cluster 4: ['Newcastle United', 'Aston Villa']
   Cluster 5: ['Manchester City', 'Chelsea', 'West Ham United']
   Cluster 6: ['Manchester United']
   Cluster 7: ['Leicester City']
   Cluster 8: ['Watford']
```

Now we need to associate tactical styles for each cluster. We are going to use the teams that are in the clusters to do so and see what each team has in common. For example, for each cluster, we need to see what the teams have in common based on the table signs (-, --, ---, +, ++, ++++) and give each cluster different playing styles. At the end, some clusters might have more than 1 factor because they play many different tactical styles. I prepared a table of what the outcomes were:

Table 6: Associating styles of play for every cluster based on the metric scores (see Appendix 2 for all the other leagues)

Cluster	Teams	Low metric score	High metric score	Associated factors

0	Bournemouth, Arsenal, Everton, Stoke City, Swansea	DefensiveActions, TimePerAction, YellowCards,	OppCounterAttackRate, Dribbles, PPDA	14, 4
1	West Bromwich Albion, Crystal Palace, Norwich, Southampton	PPDA, EffectiveTime	Crosses, AerialChallenges, WingPlayIndex, Fouls, YellowCards,	5, 7, 12
2	Liverpool, Tottenham Hotspur	LowBlockRatio ,	BuildUpSpeed, CounterattackIndex, AvgDefensiveRegainX, CentralPlayIndex, LostBalls, FinalAttemptCreation, EffectiveTime	1, 15, 13
3	Sunderland	BuildUpSpeed, OppCounterAttackRate, Crosses, Dribbles, WingPlayIndex, EffectiveTime	DefensiveActions, , SetPieceAttackRate, TimePerAction, Recoveries_OwnHalf. Fouls,	6, 11, 16
4	Newcastle United, Aston Villa	CounterattackIndex, LowBlockRatio, OppCounterAttackRate, CentralPlayIndex, FinalAttemptCreation	Recoveries_OwnHalf, LowBlockRatio	2, 9
5	Manchester City, Chelsea, West Ham United	LowBlockRatio, LostBalls,	CounterattackIndex, CentralPlayIndex, WingPlayIndex, FinalAttemptCreation	12, 3, 15
6	Manchester United	BuildUpSpeed, OppSetPieceRatio, TimePerAction	OppOpenPlayRatio, AvgDefensiveRegainX, Dribbles, EffectiveTime	9, 14, 13
7	Leicester City	PPDA	DefensiveActions, PreventedCounters_Mid3, CounterattackIndex	3, 4, 5, 10
8	Watford	CounterattackIndex	OppCounterAttackRate, YellowCards, LostBalls, Fouls, SetPieceAttackRate	6, 17

Based on the signs that each team had, I would put the metrics in the lower or higher score. For example, for clusters with several teams, I would look for signs that were the same for all the teams. Cluster 0 has 5 teams, so metrics like PPDA would be very high for all the teams, and that is one tactical trait. Another is dribbling, which every team scores all the way above average, which means that another tactical trait is individual attacking actions. Some individual clusters or other clusters would be high in a lot of metrics. In that case, I would see if they were the highest or 2nd highest in that particular metric. If that were the case, I would keep the metric; if not, I would let go, because at the end, a cluster would have too many metrics, and that makes no sense.

Some clusters may have a factor that is similar to another cluster, for example, clusters 8 and 3 both had the factor 6 (Types of attack) as one of their tactical traits, but that is normal. A lot of teams play with different tactical styles, so even though some clusters share a similar tactical trait, it makes sense and is perfectly normal.

4.6.3 Choosing the best global model

To choose the best global model, it is the same process as we saw before, we will have the clustering algorithms ranked from the best to the worst for every competition, and from there, we will choose the clustering algorithm and the number of clusters based on the clustering algorithm that is more often in the higher place in every competition.

4.7 Dimensionality reduction with PCA

PCA (Principal Component Analysis) was used on all my models for visualization purposes. There are 3 main things that PCA mostly does in my models. The first was to reduce all the dimensions of my models to 2 dimensions. All my models had 23 metrics, which have many dimensions, and the visualizations were incomprehensible. PCA can reduce high-dimensional models into standard 2 dimensions, where the visualization of the outcome makes much more sense. Another reason was to help with correlated features; some metrics were correlated, so to reduce the correlation, we can use PCA as it “transforms correlated variables into a set of uncorrelated principal components” (Medium 2024). Of course, PCA works better when we standardize the metrics before, using a mean of 0 and a standard deviation of 1, so that all metrics have the same weight (Medium 2024).

The final reason was using PCA before the clustering, by doing that, PCA will reduce the number of metrics into fewer big metrics, which we can call components. For example, my model that used 23 metrics in the Premier League was reduced to 5 components, which explained more than 80% of the tactical differences, and for the visualization, I used the first 2 components with the highest variance ratio in the 2 axes, Y and X. Here

are the 5 PCA features that PCA distinguished when I applied PCA before the clustering for the Premier League:

Style_1 (explains 54.5% variance)
Style_2 (explains 16.9% variance)
Style_3 (explains 6.6% variance)
Style_4 (explains 5.7% variance)
Style_5 (explains 4.7% variance)

Then, for the visualization in 2D, I used the first 2 styles. Here for the Premier League, the first 2 styles explained more than 70% of the variance ratio, which is not very good but still good enough to cluster teams into different tactical styles and have a 2D visualization.

5. Results

5.1 PCA interpretation for every league

Each competition had different results when PCA was applied before clustering. The Premier League had the most components, 5 in total. This is largely because in the Premier League, we can observe a large amount of tactical diversity. The Bundesliga had the fewest, 3 in total, because there were fewer matches in the Bundesliga compared to other competitions. And all the other competitions had 4 components. Here is a visualization of that:

Table 7: Results of PCA for every competition

Competition	Number of components	Total variance	Total 2D variance	Total 3D variance
Premier League	5	88.4%	71.4%	78%
Bundesliga	3	86%	78.8%	86%
La Liga	4	87.8%	72.9%	82.6%
Serie A	4	88.7%	77.4%	84.5%
Ligue 1	4	85.7%	73.3%	80.4%

From the table above, we can see that the total variances are quite similar for every competition; the differences are when we want to visualize them in different dimensions. For 2 dimensions (2D), we lose some information, which means we lose tactical information. The Premier League loses 17% of its tactical information when we reduce the dimensions to 2 dimensions (2D), which is a lot but still affordable. The Bundesliga is the one that loses the least tactical information when we reduce its dimensions because it has fewer components. Here are some takeaways from the table:

- Bundesliga requires fewer components, which indicates that there are more similar tactical styles among teams.
- The Premier League has the greatest number of components, which indicates that there is more tactical diversity in the Premier League.
- For Serie A and the Premier League, when we reduce the dimensions, we lose the most variance, which means we lose more tactical information.

- Ligue 1 and Bundesliga perform the best when we reduce the dimensions; we lose less tactical information.

5.2 League-specific best models

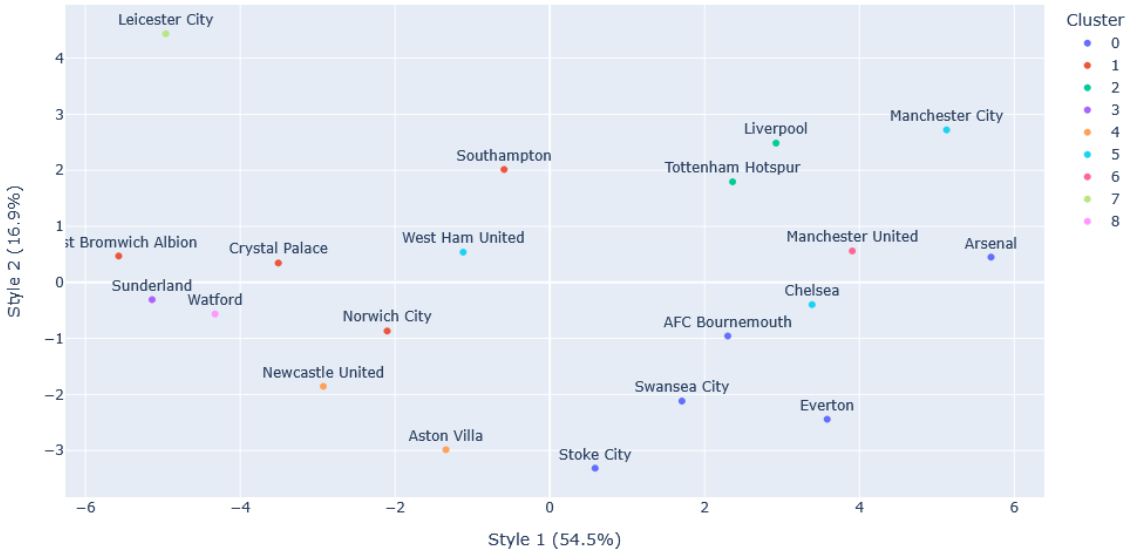
We are going to see now the best clustering algorithms per competition in two dimensions (2D).

5.2.1 Premier League

The k-means clustering algorithm with 9 clusters was the strongest and had the best overall score among all other clustering algorithms based on the evaluation metrics.

2D visualization of the Premier League:

Figure 10: 2D visualization of the Premier League



We can see that we have 9 different clusters, each cluster represents a distinct tactical style or styles of play (there can be more than one tactical style per cluster, which is always the case). Here, underneath, we can see tactical styles per cluster:

Cluster 0 (Bournemouth, Arsenal, Everton, Stoke City, Swansea): These teams had remarkably high PPDA scores, which means that they did not press a lot, so they sat down in the defense and waited. Another trait is that they had high scores in the metric OppCounterAttackRate, which means they were good at Defensive transition. Styles of play: Defensive Transition and Individual Attacking Actions.

Cluster 1 (West Bromwich Albion, Crystal Palace, Norwich, Southampton): These teams perform best in aerial challenges and crosses, had good scores in wing plays, made a lot of fouls, and had many yellow cards. This is typically a strong, rigid defensive

team that uses its wingers to cross and score. Style of play: Aerial Game, Crossing, and Width of Creative Phase.

Cluster 2 (Liverpool, Tottenham Hotspur): These were more of an attacking team, their traits are that they had a high buildUpSpeed score, which means they went to attack very quickly, and another trait is that they had high scores in FinalAttemptCreation. Style of play: Tendency to create final attempts, Effective Game, and Elaboration of Build-up Phase.

Cluster 3 (Sunderland): This team was good on set pieces and defensive actions. Style of play: Individual attacking actions, Type of Attack (set-piece attacks), and Passing Tempo.

Cluster 4 (Newcastle United, Aston Villa): These teams were good at recovering the ball in their own halves and had a rigid low block. Styles of play: Transition Game and Low Block.

Cluster 5 (Manchester City, Chelsea, West Ham United): These teams were good at counterattack, and they played a lot in the center and on the wing, so they kept the ball more than others. Styles of play: Attacking Transition, Width of Creative Phase, and Tendency to Create Final Attempts.

Cluster 6 (Manchester United): This team played a lot in defensive blocks and used a lot of individual performances. Style of play: Defensive Block, Individual Attacking Actions, and Effective Game.

Cluster 7 (Leicester City): This team was good in defensive transition, with the lowest PPDA of all, and good at aerial challenges and counterattacks. Style of play: Defensive Transition, Aerial Games, and Press.

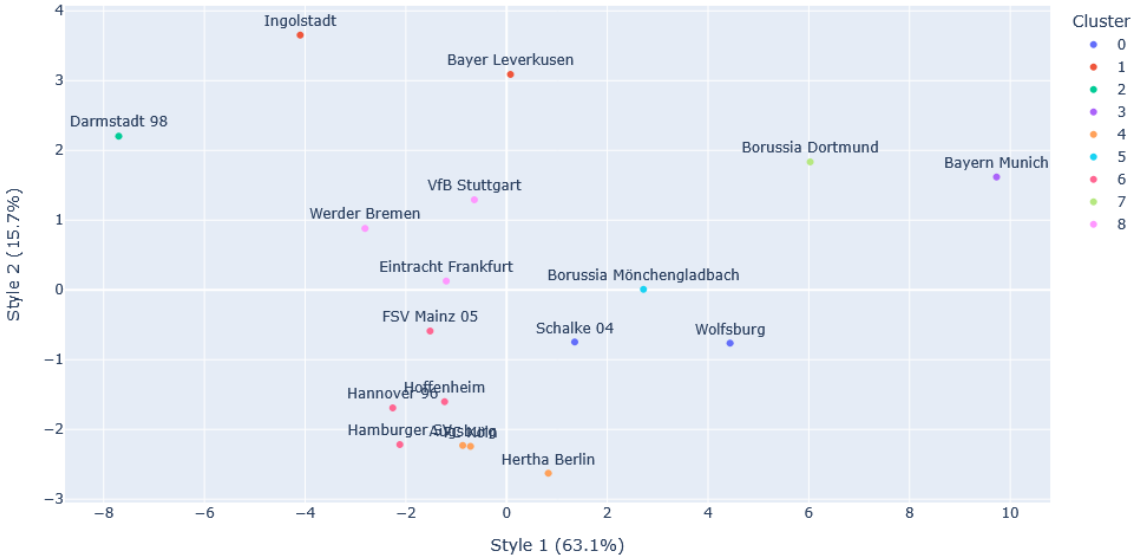
Cluster 8 (Watford): This team was good with set pieces and committed a lot of fouls and got many yellow cards. Style of play: Defending Aggressively, and Type of Attack (set-piece attacks).

5.2.2 Bundesliga

The k-means and Agglomerative clustering algorithms with 9 clusters were the strongest and had the best overall score among all other clustering algorithms based on the evaluation metrics.

2D visualization of Bundesliga using k-means:

Figure 11: 2D visualization of the Bundesliga



Cluster 0 (Schalke 04, Wolfsburg): These teams are good at creating a lot of chances and controlling the game well. Styles of play: Tendency to Create Final Attempts and Effective Game.

Cluster 1 (Bayer Leverkusen, Ingolstadt): These teams are very defensive, good at aerial games and defensive transition, and good at pressing as well. Styles of play: Aerial Game, Defensive Transition, Press, and Width of Creative Actions.

Cluster 2 (Darmstadt 98): This team defends very aggressively, makes a lot of fouls, and is bad at defending set pieces. Styles of play: Defending Aggressively, Pressing Tempo, and Press.

Cluster 3 (Bayern Munich): This team is the best in the league at attacking and building up its attacks. They play structured football and use crosses a lot. Styles of play: Crosses, Elaboration of Build-up Phases, and Type of Opponent Attack.

Cluster 4 (Hertha Berlin, Augsburg, FC Köln): These teams were a bit hard to distinguish tactically, but their biggest trait is defensive blocking. Styles of play: Defensive Block.

Cluster 5 (Borussia Mönchengladbach): This team was very good at transition games and counterattacks. Styles of play: Transition Game and Type of Attack.

Cluster 6 (Hoffenheim, Hannover 96, Hamburger SV, FSV Mainz 05): These teams excelled in transition phases and defensive transitions, as well as in breaking counters. Styles of play: Transition Game and Defensive Transition.

Cluster 7 (Borussia Dortmund): This team was the best counter-attacking team and plays well in the center and on the wing, and they control the game well. Playing styles: Effective Game, Width of Creative Phase, and Attacking Transition.

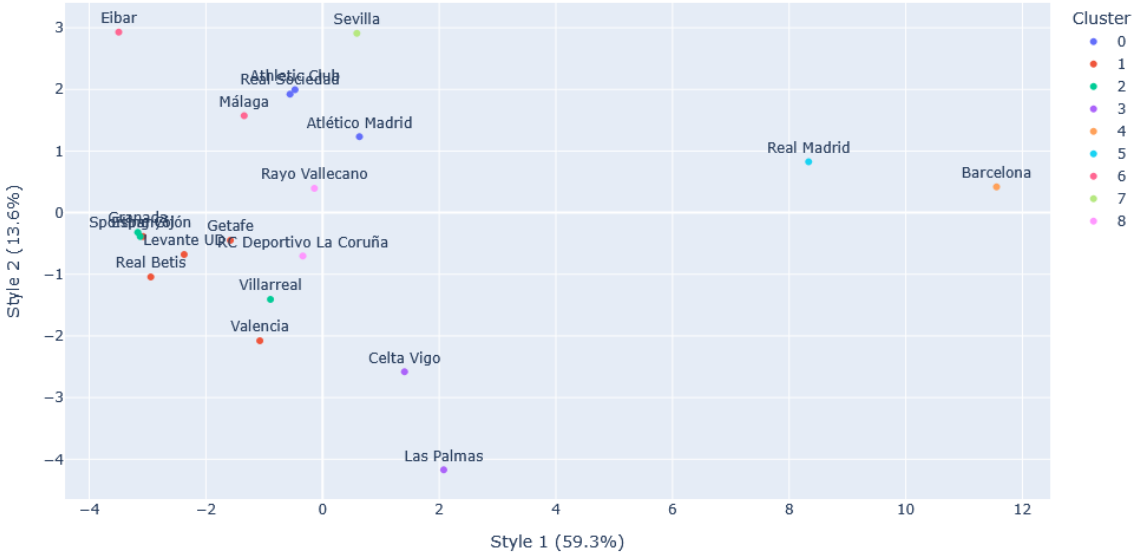
Cluster 8 (Eintracht Frankfurt, VfB Stuttgart, Werder Bremen): These teams are good at passing the ball with tempo and are good at defensive transition. Styles of play: Defensive transition and Passing tempo.

5.2.3 La Liga

The Agglomerative clustering algorithm with 9 clusters was the strongest and had the best overall score among all other clustering algorithms based on the evaluation metrics.

2D visualization of La Liga:

Figure 12: 2D visualization of La Liga



Cluster 0 (Athletic Club, Atlético Madrid, Real Sociedad): These teams have a lot of defensive traits, are good in the air, and are good at defensive transitions. They have low PPDA, so they press a lot. Styles of play: Aerial Game, Press and Crosses.

Cluster 1 (Levante UD, Getafe, Espanyol, Real Betis, Valencia): These are very defensive and aggressive teams that are good at set pieces. Styles of play: Type of Attack, Defending Aggressively, and Defensive Block.

Cluster 2 (Sporting Gijón, Villarreal, Granada): These teams press a lot and make many fouls. They also play at a high tempo. Styles of play: Press, Defending Aggressively, and Passing Tempo.

Cluster 3 (Las Palmas, Celta Vigo): These teams are possession-based, built-up teams, and have a highly effective time in building their attack. Styles of play: Elaboration of Build-up Phase, Type of Attack, and Effective Time.

Cluster 4 (Barcelona): This team never stops attacking; it creates chances, dribbles, keeps a lot of possessions, and builds up its action effectively. Fun team to watch. Styles of play: Elaboration of Build-up Phases, Individual Attacking Actions, Tendency to Create Final Attempts, and Attacking Transition.

Cluster 5 (Real Madrid): This team also attacks a lot, more on the wings and using crosses, and creates a lot of chances. Styles of play: Crosses, Width of Creative Phases, Tendency to Create Final Attempts.

Cluster 6 (Eibar, Málaga): These teams are good in the air, they press a lot, and they have a good transition game. Styles of play: Transition Game, Aerial Game, and Press.

Cluster 7 (Sevilla): This team is the best on counterattacks; they also use a lot of crosses and play more in central and on the wings. Styles of play: Width of Creative Phase, Crosses, and Attacking Transition.

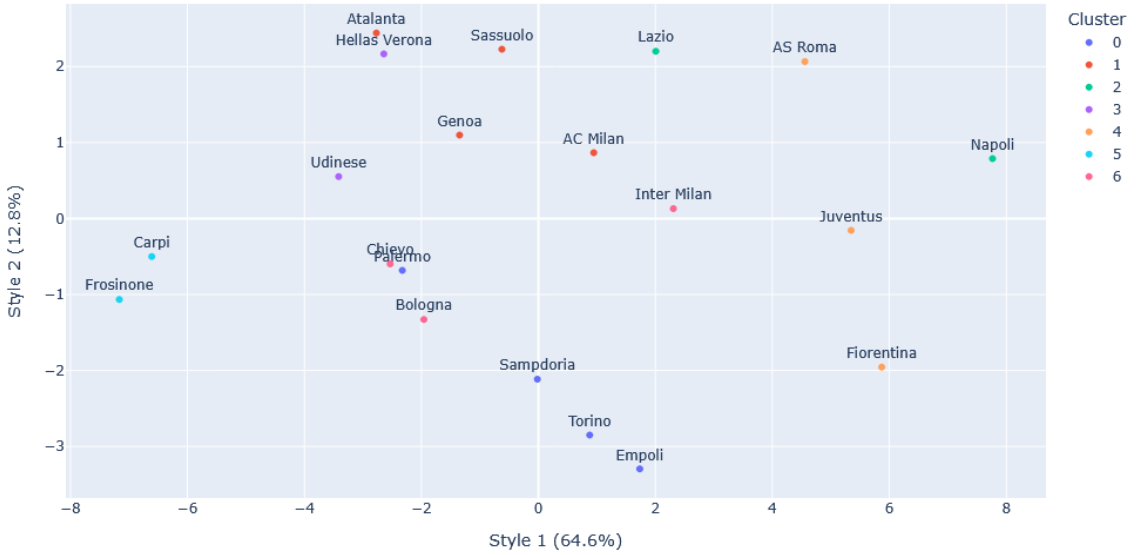
Cluster 8 (RC Deportivo La Coruña, Rayo Vallecano): These teams attack a lot, they build up fast, transition fast, and create a lot of chances. Styles of play: Tendency to Create Final Attempts and Transition Game.

5.2.4 Serie A

The Agglomerative and BIRCH clustering algorithms with 7 clusters were the strongest and had the best overall score among all other clustering algorithms based on the evaluation metrics.

2D visualization of Serie A using Agglomerative clustering:

Figure 13: 2D visualization of Serie A



Cluster 0 (Torino, Empoli, Sampdoria, Palermo): These teams played low block with a lot of individual plays and let the opponent play a lot but were good in set pieces. Styles of play: Defensive Block, Type of Opponent Play, and Individual Attacking Actions.

Cluster 1 (Atalanta, Sassuolo, AC Milan, Genoa): These teams are very dynamic and good both in defense and attack, with good counterattacks and defensive transitions. Styles of play: Defensive Transition, Attacking Transition, and Crosses.

Cluster 2 (Napoli, Lazio): These teams were well-built and created numerous chances, playing a lot in the middle and occasionally on the wings. Styles of play: Elaboration of build-up Phase, Width of Creative phase, and Tendency to Create Final Attempts.

Cluster 3 (Udinese, Hellas Verona): These were aggressive teams, with a lot of fouls and duels, and they took quick action. Styles of play: Defensive block, Defensive Transition, Crosses, and Defending Aggressively.

Cluster 4 (Fiorentina, AS Roma, Juventus): These were creative attacking teams that built up their attacks, played in the middle and on the wings. They created a lot of chances and had an effective time. Styles of play: Width of Creative Phase, Effective Time, Tendency to Create Final Attempts, and Elaboration of Build-up Phase.

Cluster 5 (Frosinone, Carpi): These teams pressed a lot, were good on set pieces, were aggressive, and stayed in a low block. Styles of play: Press, Defensive Transition, Passing Tempo, and Defending Aggressively.

Cluster 6 (Inter Milan, Bologna, Chievo): These were quite hard to determine, but they were good in the air and pressed a lot with low PPDA. Styles of play: Press and Aerial Challenges.

5.2.5 Ligue 1

The k-means clustering algorithm with 6 clusters was the strongest and had the best overall score among all other clustering algorithms based on the evaluation metrics.

2D visualization of Ligue 1:

Figure 14: 2D visualization of Ligue 1



Cluster 0 (Stade de Reims, Lorient, Saint-Étienne, Bastia, Guingamp, Angers): These teams were compact, aggressive, and had a fast tempo. Styles of play: Defending Aggressively, Passing Tempo, and Individual Defensive Actions.

Cluster 1 (Paris Saint-Germain): This was the most attacking team; they built up their attacks, created a lot of chances, and played a very controlled game. Styles of play: Elaboration of Build-up Phase, Attacking Transition, Effective Game, and Tendency to Create Final Attempts.

Cluster 2 (Rennes, Troyes, OGC Nice): These teams played more defensively and relied on individual performance more but were also effective. Styles of play: Effective Game, Individual Attacking Actions, and Defensive Block.

Cluster 3 (Nantes, Toulouse, Caen): These teams played a lot of crosses, were good on set pieces, played with a high tempo, and pressed a lot. Styles of play: Crosses, Passing Tempo, and Press.

Cluster 4 (Gazélec Ajaccio, Bordeaux): These teams pressed and played fast transition games; they played a lot of individual games. Styles of play: Press, Transition Tame, and Individual Defensive Actions.

Cluster 5 (Marseille, Lyon, AS Monaco, Montpellier, Lille): These teams were the best on the wings and played a lot in the center, with a lot of crosses and individual play. Styles of play: Crosses, Width of Creative Phase, and Individual Attacking Actions.

5.2.6 Global model

To choose the best global model, we need to choose the best clustering algorithm among the 5 competitions. Here are the rankings per competition:

Figure 16: Premier League top models

Premier League

1st K-means 9 clusters
2nd Agglomerative and Birch 8 clusters
3rd Birch 8 clusters

Figure 15: La Liga top models

La Liga

1st Agglomerative 9 clusters
2nd K-means 9 clusters
3rd GMM and Birch 8 clusters

Figure 18: Serie A top models

Serie A

1st Agglomerative and Birch 7 clusters
2nd K-means 9 clusters
3rd GMM 6 clusters

Figure 17: Bundesliga top models

Bundesliga

1st K-means and Agglomerative 9 clusters
2nd Birch 9 clusters
3rd GMM 9 clusters

Figure 19: Ligue 1 top models

Ligue 1

1st K-means 6 clusters
2nd Agglomerative 9 clusters
3rd GMM 8 clusters
4th Birch 6 clusters

Based on these rankings, k-means and Agglomerative are always in the top 2 positions in all competition rankings and are the best clustering algorithms. 9 clusters are the best out of 5 competitions. To choose one general model is very hard since every competition has its particularities, but k-means is the first in 4 out of 5 competitions, and it is the best with 9 clusters in 3 out of 5 competitions, so the best general model, if we had to choose, would be k-means with 9 clusters.

6. Discussion

6.1 Tactical alignment

To justify that my study produces good or bad outcomes, we need to see if the models for every competition align with what the football experts and professionals said about how teams played tactically in those 5 competitions in the 2015/2016 season. As I mentioned before, I was trying to create a model that would not rely on external opinions or research; rather, I was trying to create a model that would be more mathematically based using evaluation metrics, logic, and interpretation. Let's see how well my models performed based on what the external sources say.

6.2 Premier League known tactical styles of play

For the Premier League, the 2015/2016 season was particularly exciting because Leicester City won the trophy that year, which was a big surprise, even a miracle for a lot of people. In that season, Leicester City played with a well-organized defensive structure, pressed a lot, and used a lot of counterattacks. This aligns perfectly with what I found in my clusters; they played a unique style as well that season, which is why they were alone in a cluster (CE 2016a).

West Bromwich Albion, under Tony Pulis, played a lot of crosses where the goal was to find the main attacker, and they also used to play defensively with long balls up front. This also aligns with my cluster, which has crosses and aerial games as the 2 main styles of play (Braun 2016).

Liverpool that season, under Jürgen Klopp, played a new style of counter-pressing; they used to build up their attack quickly and created a lot of chances. I did have some of these traits in my cluster, but it was missing the core principle of pressing, which is a problem. But maybe the reason for that is that the counter-pressing system of Jürgen Klopp took time to be effective. In the 2015/2016 season, Klopp had just arrived, so it was tough for him to put a new system in place in the first season (EA 2016).

Manchester City under Pep Guardiola played a possession-based style; they used to have the ball a lot, attacked on the wings a lot, created a lot of chances, and were good at attacking transitions. This perfectly aligns with what I found, but they also used individual quality to attack. I did not find that in my cluster, so I was missing a tactical trait (Payne 2016a).

Tottenham Hotspur pressed a lot and played similarly to Liverpool, good in attack, used to build up their attack quickly, and created a lot of chances, but the pressing trait was

missing in my cluster. Tottenham did have low PPDA, but Liverpool did not. It is the reason why the pressing trait was not attributed to this cluster, maybe because, as I said before, this was the first season Jürgen Klopp was leading Liverpool, and this might not be the season that Liverpool started the famous counter-pressing style of Jürgen Klopp. (Payne 2016b).

Manchester United played a defensive block, their attacks relied a lot on Martial on the wings. They did control the possession of the match but were not good on the offensive. My cluster does a good job of finding tactical styles for Manchester United, but the Effective game trait might be a bit off, since Manchester that season was not very effective offensively (Davies 2016a).

Arsenal had some very talented individual players who would make the difference, like Giroud, Ozil, and Bellerin on the wings. They were also good in defensive transitions with high PPDA (lack of press). I did find all those traits in my cluster (Osmanbasic 2016a).

Newcastle United, from what I found, was the one that did not fit in my cluster; they did not play particularly in a low block, and there was no evidence of them gaining the ball in their halves more than other teams, at least this is not what the source tells us. They did play a transition game, which I found, but I only found 1 out of 3 styles of Newcastle United, which is not very good (Osmanbasic 2016a).

Crystal Palace played a lot on the wings with Zaha and Bolasie's speed and dribbling. They were very aggressive and used to make a lot of fouls. These were the traits that I found in my cluster, but the crossing from the wings tactical trait was not mentioned in the article, but they did play a lot on the wings, perhaps they crossed a lot as well (Payne 2015a).

Overall, my clusters did a good job for most of the teams, finding their different tactical styles, with some minor errors, like Newcastle United. My cluster did find some of their traits, but the problem is that they were in the same cluster as another team that might have other tactical traits that Newcastle did not have.

6.3 Bundesliga's known tactical styles of play

Borussia Dortmund, under Thomas Tuchel, dominated possession and played a lot in the middle and wide areas and controlled the game well, which aligns perfectly with my cluster (Payne 2015b).

Bayern Munich was good at building up their attacks efficiently from the back, which was one of my tactical traits that I found. They were also good at crosses, which I also found, but I didn't find wing play as one of their other tactical traits (Payne 2015c).

Ingolstadt was very aggressive in pressing and defensive transitions, aligning perfectly with what I found in the cluster that Ingolstadt was in (Payne 2015d).

Borussia Mönchengladbach was also good at defensive transitions, as this article emphasizes. I had that as one of their tactical traits in my cluster, but I couldn't find other traits (Payne 2015e).

Hertha Berlin played a structured defensive block that season, which aligns with what I found in my cluster, but I couldn't find other traits (Payne 2016c).

From external sources, most of my clusters were very good at separating teams into tactical styles of play. Still, the main problem is that I don't have information about the other teams in the same clusters. Some clusters have several teams inside them. I need to find sources that confirm that every team in those clusters had similar tactical traits. Sometimes I found one of the playing styles a team was playing, but because of a lack of more information, I could not find more tactical styles for that team.

6.4 Serie A known tactical styles of play

Juventus, that season under Massimiliano Allegri, played a 3-5-2 system that relied a lot on the wings in their build-up phase and attacked a lot as well. I did find these playing styles in my cluster (Spielverlagerung 2016a).

Roma, as well as being in the same cluster as Juventus, played a similar creative attacking style and were used to playing on the wings (Spielverlagerung 2016b). So, this cluster had both Roma and Juventus in it, and it aligns perfectly with how they played tactically that season.

Napoli was very good that season, finishing in second place. They played well-structured build-up football, played a lot on the wings, and were extremely good at attacking phases. They also scored a lot of goals. These traits align perfectly with what I found in my cluster, but I couldn't find a lot of information about Lazio that season (Davies 2016b).

Empoli dropped into a deep defensive block, did not press a lot, and let the opponent play. I found exactly those traits and a high PPDA for Empoli, which means they did not press a lot (Payne 2016d).

For Bologna, I did find that they pressed a lot, but I couldn't find the other trait, which was aerial challenges. This article does talk about how strong their defenders were and the constant passing to the striker, but no aerial statistics were given (Spielverlagerung 2016c).

For Inter Milan, which was in the same cluster, I did find the good pressing, but not the aerial challenges. So I was missing some crucial information to validate the performance of this cluster (Lawrence 2016).

AC Milan had a good defensive transition, which I found as one of their traits, but my cluster had other traits, such as attacking transition, but from what I found, they did not seem very good at attacking transitions (Thomas 2016a).

Overall, what I found was good, but there were some errors, like the Inter Milan and Bologna cluster and the AC Milan cluster, these are more a lack of information that I could not find rather than errors.

6.5 La Liga's known tactical styles of play

Barcelona was very strong that year, they had great individual players who would make the difference in big games. They played attacking football mixed with a lot of possessions and constructed build-up play. This confirms the tactical styles I found for this cluster (Payne 2015f).

Real Madrid played a lot on the wings, with Cristiano Ronaldo and Gareth Bale being very quick. Under Benitez, Real Madrid was chaotic at times and struggled with possession (Payne 2015f). In the second part of the season, a new coach, Zinedine Zidane, was appointed. He would improve Real's possessions and make them less chaotic. My cluster predicted that Real Madrid was very good on the wings and at creating chances, which was valid throughout the season, but did not find that Real Madrid became better at pressing and defensive transitions (CE 2016b).

Atletico Madrid, under Diego Simeone, is well known for its defensive style of play. In that season, they used to press a lot, use counterattacks after they intercepted and gained the ball, they used to cross a lot, and they were very strong and aggressive. This confirms perfectly what I found in that cluster (Thomas 2016b).

Valencia, another team that played a low defensive block, used a lot of set-piece threats to score goals, made a lot of fouls, and got many yellow cards. This makes my cluster very good at explaining Valencia's styles of play (Osmanbasic 2016b).

Villareal also pressed a lot and played with a high tempo, moving the ball quickly in free spaces. This aligns perfectly with what I found, but I did not find that they made a lot of fouls in this article, perhaps they did not discuss that, or it is not one of their tactical traits (Davies 2016c).

Malaga did press a lot that season, and they were good defensively. I found those styles of play, but I could not find whether they were good in the air or not (Thomas 2015).

Espanyol were very aggressive, defended deeply, and were compact. I found those styles of play but could not find if they were good at set pieces that season (Payne 2016e).

Deportivo La Coruna was a bit of a mismatch, the match that they played against Real Madrid and lost five to zero, I could not find any of the tactical traits that were in their cluster, but maybe it was because they were playing against a strong team, which made them change their styles of play for this particular match (Osmanbasic 2016c).

Overall, good results with some exceptions; most of these exceptions are more of a lack of information rather than clustering problems.

6.6 Ligue 1 known tactical styles of play

Paris Saint-Germain (PSG) had the ball a lot that season, building up their attacks patiently. They were very dangerous and flexible up front, creating a lot of dangerous situations. This perfectly aligns with what I found in my cluster (Payne 2016f).

I could not find other quality sources for other teams in Ligue 1 that talk about tactical styles of play. PSG was the only team for which I could find quality sources and data for their tactical style of play. In other sources, I could not understand how these teams played tactically; it does display their strongest playing styles, but it was very superficial, and there was no data or anything more concrete backing that up, which is the reason why I did not include them.

6.7 Conclusion regarding all competitions

Overall, we can see based on all the sources, I found that my clusters are impressively accurate in grouping teams into tactical styles of play. The only problem was that I was lacking more sources on tactical styles of play for every team, which would be more interesting if I had them, and would also help me better understand the errors my models were making.

Even if my models did a great job, there were also some errors, since I found several tactical styles per cluster, some teams did not have all these tactical traits. They were

either missing one or did not have one of these traits, like the one with Villareal and fouls. Another problem is that with Ligue 1, I have no idea if my model did a good job or not. I could only find sources for Paris Saint-Germain, for which my cluster predicted perfectly how they played that season, but to confirm that my Ligue 1 model is good and reliable, I need more information about how other teams played as well.

For all other teams that I analyzed, my models predicted their style of play almost every time, but with some minor errors, as we saw with some teams.

6.8 Limitations

6.8.1 One unified model

The primary goal of this study was to build a model that would classify teams into different tactical styles of play. That model would work on any competition in any season, including seasons other than 2015/2016. In this study, I discovered that it was not possible with my methodology. Each competition, for example, the Premier League in every season, is different, because managers and players come and go, so the tactical behaviors of the team change from one season to another. In my opinion, it is extremely difficult to create a global model that would work in all competitions. What might be more suitable is creating a global model for every competition that will work for every season that we want to study.

6.8.2 Tracking data

For this study, I was not using any tracking data because I could not get access to it for free, so I only used event data. The problem with using only event data is that we are missing some very important information during a football match. With tracking data, we can be much more precise on what is going on in a football match and see the location of every player on the pitch, which helps a lot to distinguish tactical styles based on players' movements. Another problem I had in my study is that some factors identified by (Plakias et al. 2023), like factors 18 (Attacking aggressively) and 19 (Offside trap), needed tracking data to work, so I was missing 2 very important playing styles in my study.

6.8.3 Metrics

The difficult part is choosing the best features. In my study, I wanted to use metrics as features, some more developed than others, like PPDA, which gives more insight into the tactical aspects of a team. But the quality and the precision of my metrics were not flawless. For example, the PPDA metric gives me some high scores overall, which is not what it should give. Some of my metrics are not very precise and should be worked on to improve the models. For most of the metrics, I had a hard time finding the code that

would precisely give the outcome of that specific metric; some metrics would work better if we had tracking data; those types of metrics, like the BuildUpPhase, would be much more precise if I had tracking data.

6.8.4 Concrete sources

Another important limitation of my work was the lack of sources, or more precisely, the lack of quality sources that would talk about how teams played tactically that season. I did find a good source, which was *Spielverlagerung* (Spielverlagerung 2025). Almost all of my tactical sources came from this site, but this was the only site. Having more quality sources available would be much better for my studies. Finally, I can't give a statement that says that my models are very good or not too good because I lack more concrete sources.

6.9 Future work

In the future, I suggest several works or studies that can be done from what I have already produced. 1) Improving the quality of the metrics and using all of the metrics that (Plakias et al. 2023) identified in their study. If we have all the metrics, the outcome may vary, and the results could be more positive. 2) Creating a model for a particular competition to begin with is much more effective and less time-consuming than trying to create a model for every competition. I suggest choosing a season and a competition where there is a lot of event data and tracking data available to use; this way, we can build a model with much better and more precise metrics. 3) Using my model and other studies as examples to create a classification model that would classify teams into tactical styles. This approach is more difficult because there is no clear statement of how each team plays tactically, as mentioned at the beginning of my study, so what we can do is rely on how the experts and professional football analysts define tactical styles, and from there, we can create a classification model. 4) Using tracking data to better understand tactical styles, we can also create a whole different setup of metrics. For example, with tracking data, we can create metrics only based on players' movements on the pitch, which would be interesting, because we would rely more on how players move together, in what position, and how fast in a football match. 5) Using more recent data in the model. The big problem that I had using old data was that I did not have many sources on how these teams played tactically in the 2015/2016 season. This makes it hard to give a concrete answer on how well my models worked. Using recent data will give you a clearer answer on your model's performance.

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Appendix 1: Full table of scores of every team per metric in the Premier League

Metrics 1-5

	BuildUpSpeed	AerialChallenges	CounterattackIndex	OppCounterAttackRate	LowBlockRatio
Leicester City	135.04 (--)	1903.0 (++)	2.94 (++)	9.59 (++)	23.91 (+)
AFC Bournemouth	160.76 (+)	1432.0 (-)	2.34 (-)	8.09 (+)	23.18 (-)
West Bromwich Albion	130.72 (--)	1845.0 (++)	2.48 (+)	9.89 (++)	24.3 (+)
Sunderland	136.31 (--)	1596.0 (+)	2.63 (+)	5.57 (--)	24.48 (+)
Newcastle United	144.44 (-)	1579.0 (+)	2.1 (--)	6.63 (-)	25.45 (++)
Aston Villa	147.32 (-)	1520.0 (-)	2.23 (-)	4.76 (--)	24.86 (++)
Everton	168.2 (+)	1330.0 (--)	2.23 (-)	7.68 (+)	24.63 (+)
Crystal Palace	142.94 (-)	1581.0 (+)	2.31 (-)	8.09 (+)	24.18 (+)
Watford	140.42 (-)	1663.0 (+)	1.98 (--)	10.16 (++)	24.38 (+)
Arsenal	174.29 (++)	1488.0 (-)	2.56 (+)	10.27 (++)	22.82 (-)
Liverpool	172.63 (++)	1664.0 (+)	2.7 (+)	8.57 (+)	22.72 (--)
Tottenham Hotspur	175.8 (++)	1652.0 (+)	2.87 (++)	6.84 (-)	22.79 (-)
Manchester City	171.87 (++)	1418.0 (-)	3.08 (+++)	6.76 (-)	21.97 (--)
Norwich City	144.47 (-)	1843.0 (++)	2.2 (-)	5.58 (--)	24.57 (+)
Chelsea	165.02 (+)	1350.0 (--)	2.47 (+)	6.12 (-)	23.18 (-)
Stoke City	156.68 (+)	1396.0 (--)	2.29 (-)	6.85 (-)	24.85 (++)
Manchester United	177.91 (++)	1390.0 (--)	2.43 (-)	5.36 (--)	21.68 (---)
West Ham United	146.57 (-)	1627.0 (+)	2.56 (+)	5.73 (--)	23.72 (-)
Swansea City	163.56 (+)	1462.0 (-)	2.31 (-)	8.63 (+)	24.16 (+)
Southampton	152.25 (-)	1716.0 (+)	2.56 (+)	8.64 (+)	23.19 (-)
League Average	155.36 (avg)	1572.75 (avg)	2.46 (avg)	7.49 (avg)	23.75 (avg)

Metrics 6-10

	PreventedCounters_Mid3	Crosses	OppOpenPlayRatio	OppSetPieceRatio	AvgDefensiveRegainX
Leicester City	0.56 (+++)	516.0 (+)	28.77 (--)	1.78 (++)	1117.92 (+)
AFC Bournemouth	0.39 (-)	476.0 (-)	31.07 (+)	1.58 (+)	1134.3 (+)
West Bromwich Albion	0.44 (+)	462.0 (-)	28.77 (--)	1.9 (++)	1083.7 (-)
Sunderland	0.54 (++)	329.0 (---)	29.25 (--)	1.74 (+)	1079.22 (-)
Newcastle United	0.44 (+)	475.0 (-)	29.78 (-)	1.58 (+)	1033.01 (--)
Aston Villa	0.42 (-)	486.0 (+)	30.36 (-)	1.57 (+)	1029.68 (--)
Everton	0.34 (--)	473.0 (-)	31.53 (+)	1.27 (--)	1071.26 (-)
Crystal Palace	0.51 (++)	557.0 (++)	29.32 (--)	1.83 (++)	1096.73 (-)
Watford	0.46 (+)	397.0 (--)	29.55 (-)	1.81 (++)	1094.42 (-)
Arsenal	0.33 (--)	476.0 (-)	32.02 (++)	1.2 (--)	1161.49 (+)
Liverpool	0.38 (-)	511.0 (+)	31.38 (+)	1.35 (-)	1189.63 (++)
Tottenham Hotspur	0.41 (-)	431.0 (-)	31.24 (+)	1.5 (-)	1156.66 (+)
Manchester City	0.39 (-)	540.0 (++)	31.84 (++)	1.31 (--)	1223.99 (++)
Norwich City	0.4 (-)	507.0 (+)	30.15 (-)	1.67 (+)	1078.23 (-)
Chelsea	0.37 (-)	459.0 (-)	31.61 (+)	1.37 (-)	1162.03 (+)
Stoke City	0.39 (-)	404.0 (--)	30.86 (+)	1.38 (-)	1066.88 (-)
Manchester United	0.43 (+)	521.0 (+)	31.86 (++)	1.26 (--)	1242.41 (+++)
West Ham United	0.44 (+)	477.0 (-)	29.93 (-)	1.7 (+)	1105.39 (-)
Swansea City	0.38 (-)	442.0 (-)	31.37 (+)	1.41 (-)	1094.31 (-)
Southampton	0.46 (+)	606.0 (+++)	30.5 (-)	1.53 (-)	1133.38 (+)
League Average	0.42 (avg)	477.25 (avg)	30.56 (avg)	1.54 (avg)	1117.73 (avg)

Metrics 11-15

	PPDA	DefensiveActions	CentralPlayIndex	WingPlayIndex	Dribbles
Leicester City	663.13 (---)	2785.0 (+++)	0.15 (++)	0.14 (++)	13.42 (--)
AFC Bournemouth	1169.24 (+)	2238.0 (--)	0.11 (-)	0.13 (+)	14.97 (+)
West Bromwich Albion	764.82 (--)	2515.0 (+)	0.1 (-)	0.12 (+)	13.48 (--)
Sunderland	860.46 (--)	2623.0 (++)	0.12 (-)	0.1 (--)	13.7 (--)
Newcastle United	1141.37 (+)	2709.0 (++)	0.1 (--)	0.12 (+)	14.25 (-)
Aston Villa	1139.67 (+)	2414.0 (-)	0.09 (--)	0.11 (-)	14.66 (+)
Everton	1308.68 (++)	2314.0 (--)	0.13 (+)	0.12 (-)	15.38 (++)
Crystal Palace	822.62 (--)	2448.0 (-)	0.1 (--)	0.13 (+)	14.15 (-)
Watford	921.48 (-)	2550.0 (+)	0.12 (-)	0.1 (--)	13.73 (--)
Arsenal	1287.69 (++)	2312.0 (--)	0.16 (++)	0.13 (+)	15.47 (++)
Liverpool	1137.93 (+)	2560.0 (+)	0.14 (+)	0.12 (+)	14.99 (+)
Tottenham Hotspur	1038.17 (-)	2451.0 (-)	0.15 (++)	0.11 (-)	15.08 (+)
Manchester City	1178.57 (+)	2424.0 (-)	0.15 (++)	0.15 (+++)	15.49 (++)
Norwich City	1171.51 (+)	2391.0 (-)	0.1 (--)	0.14 (++)	14.12 (-)
Chelsea	1156.65 (+)	2403.0 (-)	0.15 (++)	0.12 (+)	15.03 (+)
Stoke City	1182.83 (+)	2356.0 (-)	0.11 (-)	0.1 (--)	14.79 (+)
Manchester United	1137.84 (+)	2411.0 (-)	0.13 (+)	0.13 (+)	15.31 (++)
West Ham United	900.91 (-)	2478.0 (+)	0.13 (+)	0.12 (+)	14.54 (-)
Swansea City	1299.47 (++)	2392.0 (-)	0.1 (-)	0.12 (-)	15.01 (+)
Southampton	974.95 (-)	2508.0 (+)	0.14 (+)	0.13 (+)	14.41 (-)
League Average	1062.90 (avg)	2464.10 (avg)	0.12 (avg)	0.12 (avg)	14.60 (avg)

Metrics 16-20

	TimePerAction	LostBalls	Recoveries_OwnHalf	Fouls	YellowCards
Leicester City	139.74 (++)	3736.0 (++)	2296.0 (++)	0.57 (+)	0.05 (+)
AFC Bournemouth	109.17 (-)	3619.0 (+)	1843.0 (--)	0.42 (-)	0.04 (--)
West Bromwich Albion	150.72 (++)	3512.0 (-)	2115.0 (+)	0.63 (++)	0.07 (++)
Sunderland	143.45 (++)	3688.0 (+)	2218.0 (++)	0.61 (++)	0.07 (++)
Newcastle United	123.81 (+)	3457.0 (-)	2330.0 (+++)	0.55 (+)	0.06 (+)
Aston Villa	120.39 (+)	3416.0 (-)	2078.0 (+)	0.54 (+)	0.06 (+)
Everton	105.56 (-)	3356.0 (-)	1951.0 (-)	0.37 (--)	0.04 (--)
Crystal Palace	130.83 (+)	3489.0 (-)	2030.0 (-)	0.66 (++)	0.07 (+)
Watford	131.46 (+)	3778.0 (++)	2114.0 (+)	0.62 (++)	0.07 (++)
Arsenal	94.02 (--)	3306.0 (--)	1891.0 (--)	0.35 (--)	0.03 (---)
Liverpool	98.17 (--)	3829.0 (++)	2084.0 (+)	0.43 (-)	0.05 (-)
Tottenham Hotspur	100.43 (-)	3706.0 (++)	1980.0 (-)	0.46 (-)	0.06 (+)
Manchester City	93.18 (--)	3422.0 (-)	1933.0 (-)	0.41 (-)	0.05 (-)
Norwich City	129.12 (+)	3755.0 (++)	2031.0 (-)	0.56 (+)	0.07 (++)
Chelsea	101.03 (-)	3273.0 (--)	1965.0 (-)	0.42 (-)	0.04 (--)
Stoke City	117.23 (+)	3355.0 (-)	2007.0 (-)	0.52 (+)	0.05 (-)
Manchester United	96.68 (--)	3559.0 (+)	1946.0 (-)	0.47 (-)	0.06 (+)
West Ham United	125.22 (+)	3287.0 (--)	2078.0 (+)	0.49 (-)	0.05 (-)
Swansea City	106.13 (-)	3465.0 (-)	1993.0 (-)	0.44 (-)	0.05 (-)
Southampton	119.01 (+)	3585.0 (+)	2064.0 (+)	0.55 (+)	0.06 (+)
League Average	116.77 (avg)	3529.65 (avg)	2047.35 (avg)	0.50 (avg)	0.05 (avg)

Metrics 21-23

	FinalAttemptCreation	SetPieceAttackRate	EffectiveTime
Leicester City	3.63 (-)	0.97 (++)	51786.66 (--)
AFC Bournemouth	3.62 (-)	0.8 (-)	65729.79 (+)
West Bromwich Albion	2.78 (--)	1.07 (++)	48887.24 (--)
Sunderland	3.12 (-)	0.96 (++)	50281.85 (--)
Newcastle United	3.01 (-)	0.83 (-)	57708.52 (-)
Aston Villa	2.94 (--)	0.8 (-)	59692.96 (-)
Everton	3.87 (+)	0.75 (-)	68947.0 (+)
Crystal Palace	3.44 (-)	1.01 (++)	55244.24 (-)
Watford	3.2 (-)	1.0 (++)	54730.83 (-)
Arsenal	4.73 (++)	0.73 (-)	73566.2 (++)
Liverpool	4.9 (++)	0.72 (-)	73599.71 (++)
Tottenham Hotspur	4.82 (++)	0.77 (-)	71473.94 (++)
Manchester City	4.89 (++)	0.7 (--)	73338.01 (++)
Norwich City	3.14 (-)	0.82 (-)	56597.07 (-)
Chelsea	4.04 (+)	0.83 (-)	70191.76 (+)
Stoke City	3.07 (-)	0.72 (-)	60567.39 (-)
Manchester United	3.25 (-)	0.65 (--)	74373.8 (++)
West Ham United	4.09 (+)	1.0 (++)	58249.66 (-)
Swansea City	3.36 (-)	0.77 (-)	65826.34 (+)
Southampton	3.86 (+)	0.82 (-)	59704.82 (-)
League Average	3.69 (avg)	0.84 (avg)	62524.89 (avg)

Appendix 2: Associating styles of play for every cluster based on the metric scores for every league

Bundesliga :

Cluster	Teams	Low metric score	High Metric score	Associated Factors
0	Schalke 04, Wolfsburg	Recoveries_OwnHalf, DefensiveActions, AerialChallenges	EffectiveTime , FinalAttemptCreation,Dribbles, PPDA,OppOpenPlayRatio, BuildUpSpeed	15, 13
1	Bayer Leverkusen, Ingolstadt	LowBlockRatio, PPDA	AerialChallenges, PreventedCounters_Mid3, AvgDefensiveRegainX, WingPlayIndex, LostBalls	4, 5, 10, 12
2	Darmstadt 98	Dribbles, EffectiveTime	OppSetPieceRatio, TimePerAction YellowCards, Fouls, SetPieceAttackRate	17, 16, 10
3	Bayern Munich	LowBlockRatio, PreventedCounters_Mid3,	BuildUpSpeed,Crosses, AvgDefensiveRegainX, PPDA, CentralPlayIndex FinalAttemptCreation, EffectiveTime	1, 7, 8
4	Hertha Berlin, Augsburg, FC Köln	AvgDefensiveRegainX,	LowBlockRatio, PPDA	9
5	Borussia Mönchengladbach	Crosses, OppCounterAttackRate, YellowCards	OppOpenPlayRatio, CounterattackIndex	3, 8
6	Hoffenheim, Hannover 96,	Crosses, AvgDefensiveRegainX, CentralPlayIndex	OppCounterAttackRate LowBlockRatio, PreventedCounters_Mid3,	2, 4

	Hamburger SV, FSV Mainz 05		LostBalls, Recoveries_OwnHalf, Fouls	
7	Borussia Dortmund	Fouls, YellowCards	CentralPlayIndex, PPDA, CounterattackIndex, Dribbles	3, 12, 13
8	Eintracht Frankfurt, VfB Stuttgart, Werder Bremen	CounterattackIndex	PreventedCounters_Mid3, OppSetPieceRatio, TimePerAction Crosses, YellowYards	4,16

La Liga :

Cluster	**Teams**	**Low metric score**	**High Metric score**	**Associated Factors**
0	Athletic Club, Atlético Madrid, Real Sociedad	LowBlockRatio, PPDA	AerialChallenges, OppCounterAttackRate, PreventedCounters_Mid3,Crosses, AvgDefensiveRegainX, WingPlayIndex, LostBalls	5, 7, 10
1	Levante UD, Getafe, Espanyol, Real Betis, Valencia		LowBlockRatio, OppSetPieceRatio,Fouls, YellowCrad, SetPieceAttackRate	17, 9, 6
2	Sporting Gijón, Villarreal, Granada	BuildUpSpeed, OppCounterAttackRate, PPDA	TimePerAction,Fouls, YellowCrad,	17, 16, 10
3	Las Palmas, Celta Vigo	Crosses, WingPlayIndex	BuildUpSpeed, OppCounterAttackRate, OppOpenPlayRatio,Dribbles, Recoveries_OwnHalf, EffectiveTime	1, 8, 13

4	Barcelona	OppSetPieceRatio ,	BuildUpSpeed, CounterattackIndex. OppOpenPlayRatio,PPDA, CentralPlayIndex,Dribbles, FinalAttemptCreation, EffectiveTime	1, 8, 3, 14, 13, 15
5	Real Madrid	LostBalls	Crosses,FinalAttemptCreation, FinalAttemptCreation BuildUpSpeed,WingPlayIndex, CentralPlayIndex	8, 1, 7, 15, 12
6	Eibar, Málaga	Dribbles, PPDA	AerialChallenges, PreventedCounters_Mid3, AvgDefensiveRegainX,LostBalls, Fouls	5, 10, 2
7	Sevilla	Recoveries_OwnHalf	CounterattackIndex,Crosses, WingPlayIndex	3, 7, 12
8	RC Deportivo La Coruña, Rayo Vallecano	OppCounterAttackRate	LostBalls, FinalAttemptCreation,BuildUpSpeed, LowBlockRatio	15, 2

Serie A :

Cluster	**Teams**	**Low metric score**	**High Metric score**	**Associated Factors**
0	Torino, Empoli, Sampdoria, Palermo		LowBlockRatio, OppOpenPlayRatio,PPDA , Dribbles	9, 8, 14
1	Atalanta, Sassuolo, AC Milan, Genoa		CounterattackIndex, PreventedCounters_Mid3, Crosses	3, 7, 4, 6
2	Napoli, Lazio	SetPieceAttackRate, LowBlockRatio	BuildUpSpeed, AvgDefensiveRegainX, CentralPlayIndex, FinalAttemptCreation	1, 12, 15

3	Udinese, Hellas Verona	EffectiveTime	AerialChallenges, OppCounterAttackRate, LowBlockRatio,Crosses, TimePerAction,Fouls, YellowCards	4, 17, 7, 9
4	Fiorentina, AS Roma, Juventus	AerialChallenges	PPDA,FinalAttemptCreation, BuildUpSpeed, CentralPlayIndex, , WingPlayIndex, FinalAttemptCreation, EffectiveTime	1, 12, 13, 15
5	Frosinone, Carpi	BuildUpSpeed,PPDA	LowBlockRatio, PreventedCounters_Mid3, OppSetPieceRatio, DefensiveActions, TimePerAction, Fouls YellowCards, SetPieceAttackRate	10, 16, 17 ,4
6	Inter Milan, Bologna, Chievo	FinalAttemptCreation, PPDA	AerialChallenges	5, 10

Ligue 1 :

Cluster	Teams	Low metric score	High Metric score	Associated Factors
0	Stade de Reims, Lorient, Saint-Étienne, Bastia, Guingamp, Angers	OppOpenPlayRatio	YellowCards,Fouls , LowBlockRatio, OppSetPieceRatio, DefensiveActions TimePerAction	17, 11, 16
1	Paris Saint-Germain	LowBlockRatio, LostBalls, Recoveries_OwnHalf, Fouls, YellowCards	BuildUpSpeed, CounterattackIndex, OppOpenPlayRatio, PPDA, FinalAttemptCreation, EffectiveTime	1, 3, 13, 15

2	Rennes, Troyes, OGC Nice	SetPieceAttackRate, OppSetPieceRatio	LowBlockRatio, PPDA, Dribbles, EffectiveTime	13, 14, 9
3	Nantes, Toulouse, Caen	PPDA	AerialChallenges, PreventedCounters_Mid3, Crosses, CounterattackIndex, DefensiveActions, TimePerAction, WingPlayIndex, Recoveries_OwnHalf	7, 16, 10
4	Gazélec Ajaccio, Bordeaux	Dribbles,PPDA	Recoveries_OwnHalf, DefensiveActions,	10, 2, 11
5	Marseille, Lyon, AS Monaco, Montpellier, Lille	Recoveries_OwnHalf	CentralPlayIndex, OppCounterAttackRate, WingPlayIndex, Crosses, AvgDefensiveRegainX, Dribbles	12, 7, 14

Appendix 3: GitHub code link

Code in the following link: https://github.com/Beniamin212/Tactical-styles

